Probabilistic Machine Learning

Lecture 03: Generative Models and Naive Bayes

Prof. Dr. Michael Granitzer

Speaker notes

Five modules. Three kinds of classifier: a rule maps x to a class without any probability; a discriminative classifier models $P(y \mid x; \theta)$ only; a generative classifier models $P(y)$ and $p(x \mid y)$, obtains the posterior by Bayes' theorem and can generate data. The curse of dimensionality: read as a sample, the training table cannot fill a full $p(x \mid y)$, whose cells multiply with every attribute (the curse of dimensionality), so the model must be restricted — by a simpler structure, by parametric families, or both. Naive Bayes: conditional independence of the attributes given the class, which reduces the number of parameters from $2^p - 1$ to p per class for p binary attributes; fitting is counting, and a count of zero vetoes all other evidence unless it is smoothed. Continuous attributes: a density per class, of which Bayes' theorem needs only the ratio; Gaussian naive Bayes; one Gaussian per class keeps the correlations at a cost quadratic in p; a shared covariance makes the boundary linear, the logistic form of Lecture 04; a class with two patterns needs a mixture (Lecture 07). Deciding on a class: the most probable class minimises the error rate (the Bayes decision rule); unequal costs move the threshold on the posterior to $c_{FP}/(c_{FP} + c_{FN})$; for a generative model prior and costs meet in one threshold on the likelihood ratio.

Three choices are independent of each other: what to model (generative or discriminative), how to fit its parameters, and how to decide on a class. This lecture fits by counting and by class means and spreads; Lecture 04 opens by showing what these estimates are (the peak of a posterior over the parameter, the maximum likelihood estimate; +1 smoothing is its mean) and then fits discriminative models the same way. Fraud detection (base rate 1 in 1,000, red flags and the amount) is the running example.

Where We Left Off

Lecture 02: 100 students, a count table.

	PML	DB	total
fail	10	6	16
pass	30	54	84
total	40	60	100

$$P(\text{PML} \mid \text{fail}) = \frac{P(\text{fail} \mid \text{PML})\,P(\text{PML})}{P(\text{fail})} = \frac{10}{16}$$

One hypothesis, **one** piece of evidence: Bayes' theorem reads the table backwards.

Speaker notes

The count table of Lecture 02 is a joint distribution of two variables, course and result, given as counts. Bayes' theorem read it backwards: the prior $P(\mathrm{PML}) = 40/100$, the likelihood $P(\mathrm{fail} \mid \mathrm{PML}) = 10/40$, the evidence $P(\mathrm{fail}) = 16/100$, so $P(\mathrm{PML} \mid \mathrm{fail}) = 10/16 \approx 0.63$.

A generative classifier is exactly this computation with the course replaced by the class y and the exam result by an attribute vector $x = (x_1, \ldots, x_p)$: the class prior $P(y)$ and the class-conditional distribution $p(x \mid y)$ are estimated from a labelled table and combined by Bayes' theorem.

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> **One** hypothesis, **one** piece of evidence: Bayes' theorem reads the table backwards.

Now the hypothesis is a class — fraud or honest — and the evidence a whole transaction. What changes?

From One Attribute to Many

Speaker notes

The four questions are the extensions this lecture takes up. Several variables: $p(x \mid y)$ is a joint distribution over all attributes, with one number per combination of values ($2^{30} - 1$ per class for 30 binary flags). Dependence: the joint is not determined by its margins, so each dependence is a number of its own; naive Bayes assumes them away. Numbers: a continuous attribute has a density, not a probability per value, so it needs bins or a parametric family. Probabilities at all: a rule maps x to a class without any distribution (a discriminant function); logistic regression models $p(y \mid x)$ only (Lecture 04); the generative route models $p(x, y)$.

From One Attribute to Many

Many variables at once. Amount, country, hour, device — how do they combine into one probability per class?

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Numbers, not yes/no. What is the probability of *1,250.00 €*?

From One Attribute to Many

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Dependent variables. Does a new device make a foreign IP more likely?

Numbers, not yes/no. What is the probability of *1,250.00 €*?

Probabilities at all? A rule *if amount > 1k € and new device, then fraud* also classifies.

Today: Generative Classifiers and the Bayes Decision Rule

question	module	the answer, in short
Probabilities at all?	**Three Kinds of Classifier**	a rule gives a class; a model of $p(x, y)$ gives $P(y \mid x)$, $p(x)$ and new data — if the joint can be estimated
Many variables at once?	**The Curse of Dimensionality**	the joint explodes; two ways out — simpler structure, a family
Dependent variables?	**Naive Bayes**	assume them independent given the class: one small table each
Numbers, not yes/no?	**Continuous Attributes**	a density per class; one Gaussian keeps the dependences
Which class, then?	**Deciding on a Class**	the most probable one — or a threshold set by costs and prior

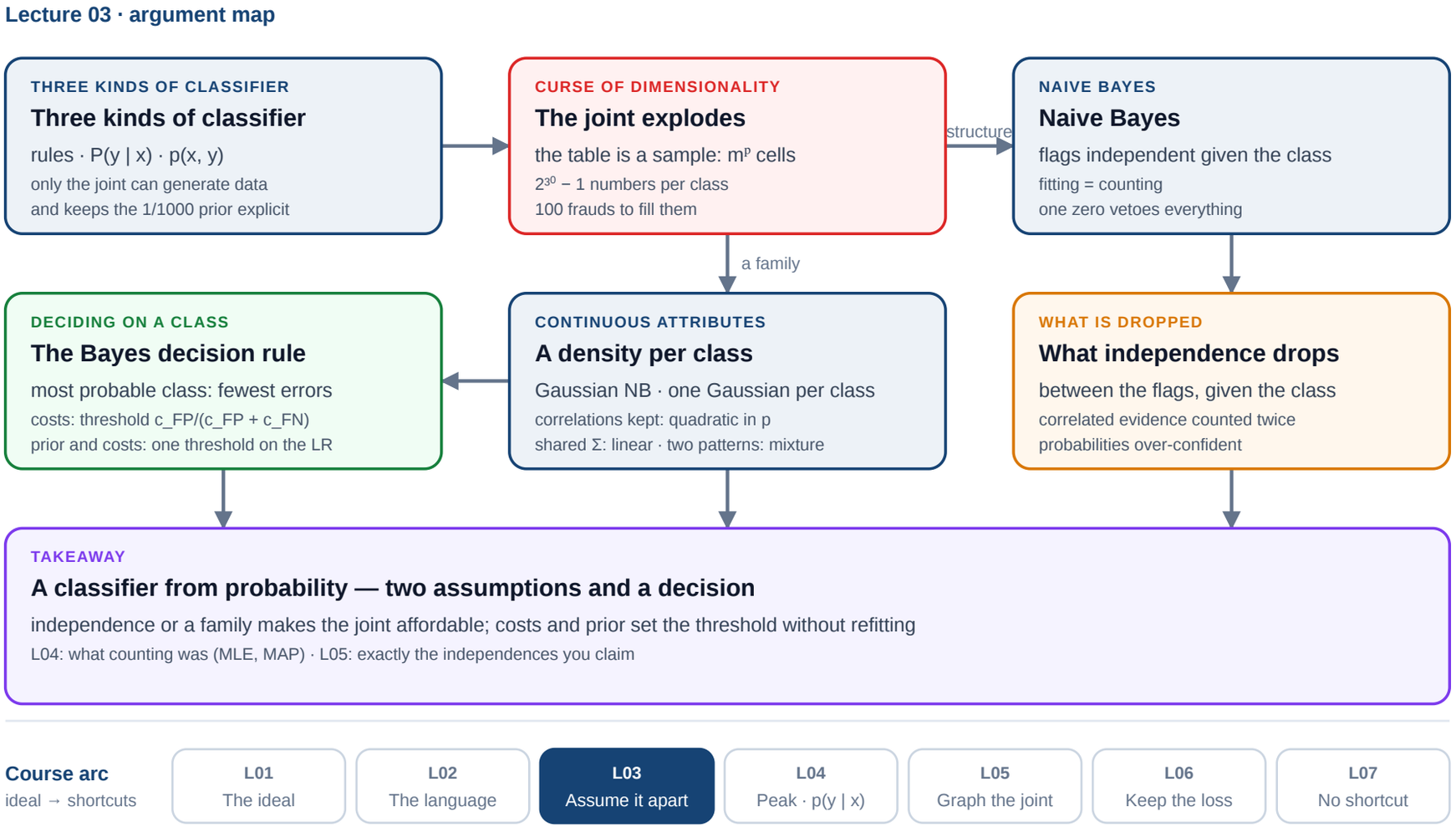
Speaker notes

The rows follow the questions of the previous slide, with the decision added: a posterior is not yet an action. Two assumptions make the joint estimable, the two ways out of the curse of dimensionality. Conditional independence given the class is the shortcut "assume it apart" of Lecture 01's table: it replaces $p(x \mid y)$ by $\prod_j p(x_j \mid y)$ and discards the dependences between attributes. A parametric family replaces a table of counts by a formula with few parameters, a normal for an amount; one normal per class keeps the dependences at a cost that grows quadratically, not exponentially, with the number of attributes, but it cannot follow a class with several patterns.

A generative classifier as such requires neither assumption; they are what makes it affordable. How the parameters are fitted — by counting, by class means and spreads — is a choice of its own, made in the same way for discriminative models; Lecture 04 opens with it. The decision is a third, separate choice: the Bayes decision rule for equal costs, a threshold from the costs otherwise.

Running example: fraud detection with a base rate of 1 in 1,000 (prior odds 1 : 999), a handful of binary red flags and the amount.

Argument Map



Speaker notes

The map follows the five modules. Of the three kinds of classifier, only the generative one models $p(y)$ and $p(x \mid y)$; it converts with Bayes' theorem, keeps the 1/1,000 prior explicit and can generate data. Its problem is the joint: read as a sample, the table puts almost no data into each of the m^p cells ($2^{30} - 1$ numbers per class for 30 binary flags, against 100 frauds). There are two ways out. Structure: naive Bayes assumes the flags independent given the class, so fitting becomes counting; it drops the dependences between flags, so correlated evidence is counted twice and the probabilities are over-confident. A family: a density per class for each number, Gaussian naive Bayes, or one Gaussian per class, which keeps the correlations at a cost quadratic in the number of attributes; a shared covariance makes the boundary linear, and a class with two patterns needs a mixture. The Bayes decision rule turns the posterior into a class: the most probable class makes the fewest errors, unequal costs move the threshold, and prior and costs enter as one threshold on the likelihood ratio, so both change without refitting.

What the counts and class means were — the peak of a posterior over the parameter, the maximum likelihood estimate — opens L04, which fits discriminative models the same way; L05 states exactly which independences a model assumes. The course-arc strip places the lecture after the language of probability (L02).

Three Kinds of Classifier

Guiding Question: A rule, a model of $P(y \mid x)$, a model of how the data arise — what can each one tell us about a transaction, and what does the most informative one cost?

Data and Hypothesis as Random Variables

amount	distance	…	fraud?
49.90 €	3 km	…	no
1,250 €	900 km	…	**yes**
X_1	X_2	…	Y

- **Data** $x = (x_1, \ldots, x_p)$: the values of the attributes $X_1, \ldots, X_p$ — what we observe.
- **Hypothesis** y: the value of the class variable Y — what we want to know.
- **Decision:** the posterior $P(y \mid x)$.

Speaker notes

This fixes the notation for the rest of the course. Each column of a labelled table is a random variable, as in Lecture 02; the attribute columns are $X_1, \ldots, X_p$ and the label column is Y. A row is one realisation (x, y) of the vector $(X_1, \ldots, X_p, Y)$, and the table is a sample of n such draws.

The names follow the reading of Bayes' theorem: y is the hypothesis (which class), x the data (what was observed about the case). The count table of Lecture 02 has the course as the hypothesis, y, and the exam result as the data, x: $P(\text{PML} \mid \text{fail})$ is $P(y \mid x)$. The textbook letters vary (Bishop: $\mathbf{x}$ and t or $\mathcal{C}_k$; MacKay: D and H); in machine learning x and y are standard. Capital letters denote the random variables, small letters their values; x is a vector whenever there is more than one attribute. In general statements the attributes are numbered, $x_1, \ldots, x_p$; in the examples they are indexed by name, x_{amt} for the amount and x_{dist} for the distance.

Data and Hypothesis as Random Variables

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- **Decision:** the posterior $P(y \mid x)$.

The student table had hypothesis A = "PML" and evidence B = "fail". Which of the two is the data, which the hypothesis?

Parameters θ

A **model** is a family of distributions $p(\,\cdot\,;\theta)$. The **parameters** θ pick one member; training sets them.

Read the notation

$p(x \mid y;\theta)$ — the probability of data x, **given** the hypothesis y, **for** the parameter value θ.

The bar conditions on a random variable, the semicolon fixes a parameter.

Example

Fraud share: $P(Y = \text{fraud};\theta) = \pi$, one number.

Log amount per class ~ normal: $\mu_{\text{f}}, \sigma_{\text{f}}$ and $\mu_{\text{h}}, \sigma_{\text{h}}$.

Together: $\theta = (\pi, \mu_{\text{f}}, \sigma_{\text{f}}, \mu_{\text{h}}, \sigma_{\text{h}})$.

Speaker notes

Lecture 02 wrote a parametric family as $p(x \mid \theta)$. From here on the parameters are separated by a semicolon, $p(x; \theta)$ and $p(x \mid y; \theta)$, so that the bar is reserved for conditioning on random variables of the model (data and hypothesis). The two notations mean the same density; the semicolon says that θ is treated as a fixed, unknown number, not as a random variable.

θ always stands for all parameters of the model; its components get names of their own — π for the class share (Bishop's notation), μ and σ per class for a normal. A model with a single parameter, such as the Bernoulli model of Lecture 04, writes θ for that one number. The amount is modelled on a log scale, as the module on continuous attributes shows: amounts are skewed, their logarithms close to normal.

The bar returns for θ once θ gets a distribution of its own: a prior $p(\theta)$ and a posterior $p(\theta \mid D)$ over the parameters given the training data D, as in Lecture 04. Every model in this course has parameters in this sense — the thresholds of a rule, the weights of logistic regression, the means and spreads of class distributions.

Rule-Based Classifier

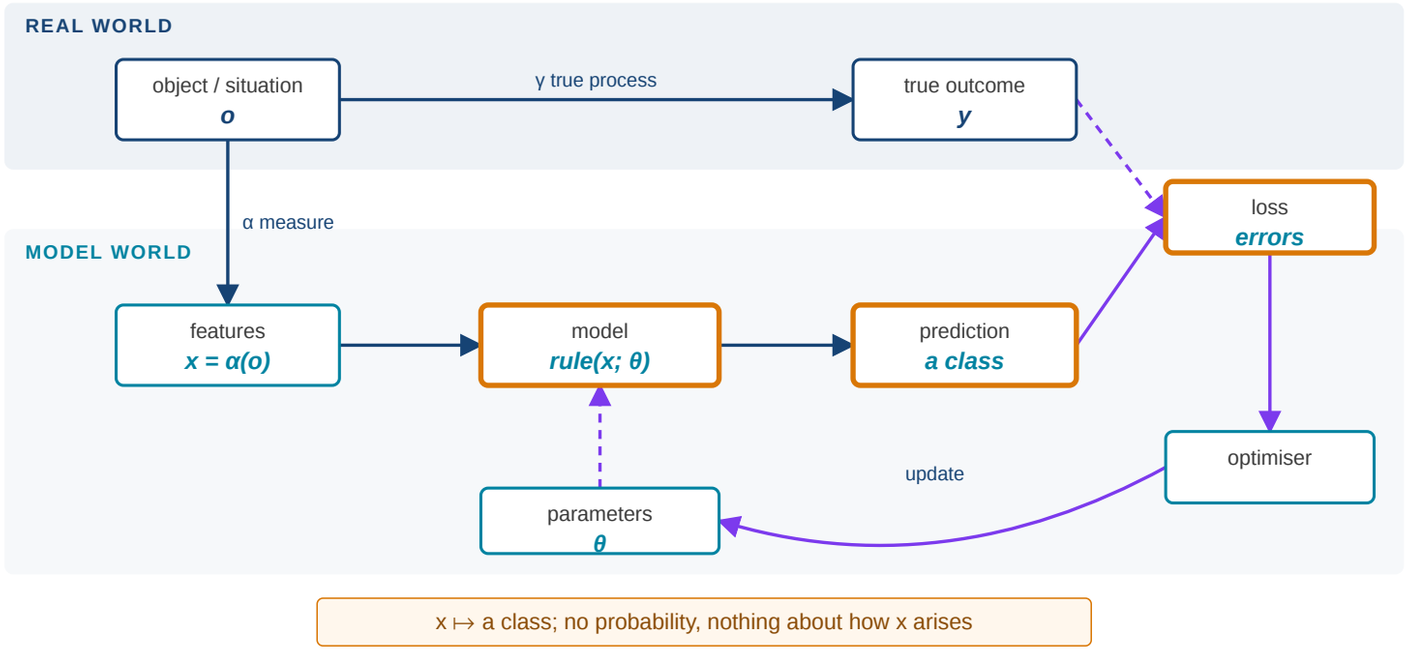
A **rule-based classifier**: rules "if $C_k(x)$ then c_k" plus a default class. The **rules are the parameters θ**.

Example: $x_{\text{amt}} > a$ and $x_{\text{dist}} > b \rightarrow$ fraud, else honest; $\theta = (a, b)$. **Training:** the fewest errors on the n rows,

$$\hat{\theta} = \arg\min_{\theta} \sum_{i=1}^{n} 1[\hat{y}_i \neq y_i]$$

- y_i: label of row i; $\hat{y}_i = \text{rule}(x_i; \theta)$: the rule's class
- $1[\cdot]$: 1 if true, else 0 — the sum counts errors
- $\arg\min_\theta$: the θ with the fewest; $\hat{\theta}$: the fitted θ

Discriminant function: rules



x ↦ a class; no probability, nothing about how x arises

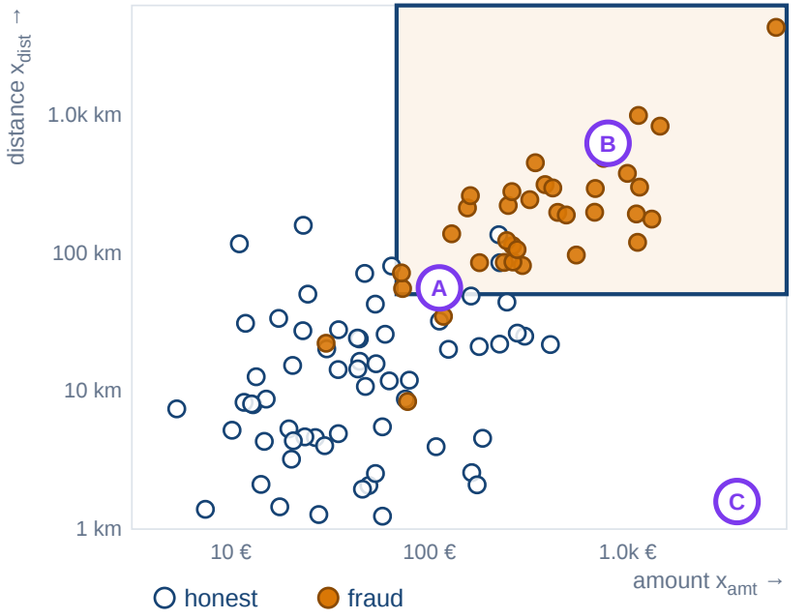
Speaker notes

The learning system of the introductory lecture with a rule learner in the model box. The features are the attributes x; the model computes $\mathrm{rule}(x;\theta)$; the prediction is a class; the loss counts errors on the training rows (the 0/1 loss, $1[\hat{y}_i \neq y_i]$ with $\hat{y}_i = \mathrm{rule}(x_i;\theta)$), and the optimiser searches over θ. The parameters of a rule set are its structure and its thresholds: which attributes the conditions test, the cut points a, b, the class of each rule and their order (first match wins). For a fixed form, as in the example, only the thresholds are free; a rule learner (RIPPER, CN2) or a decision tree (CART, C4.5) also chooses the conditions, greedily, because the number of possible rule sets grows combinatorially. A decision tree is a rule set whose conditions are the paths from the root to the leaves.

No component of the system represents a probability: there is nothing to condition on, nothing to marginalise and no prior to change. The 0/1 loss is piecewise constant in θ, so it is minimised by search over candidate thresholds, not by gradients.

Rules From Data

A and B get the same answer; C gets one without any warning



Learned rule

if amount > 68 €
and distance > 51 km
then fraud, else honest

training errors: 5 of 98

fraud share in the box: 94%
— the only "score" the rule has

A: fraud, 94%
just inside the box

B: fraud, 94%
deep inside the box

C: honest
3.5k € at 1.6 km — no such row

Which transaction is the rule surest about — A, B or C?

Speaker notes

The data are 98 labelled transactions with two attributes, the amount x_{amt} and the distance from the card holder's home x_{dist} (both on log axes); fraud is over-represented (a third) so that it is visible. The rule is learned from the table: among all rules of the form "amount $> a$ and distance $> b \rightarrow$ fraud", the thresholds (a, b) with the fewest training errors are chosen by trying every split between neighbouring values — the rule and the training criterion of the previous slide. The learned box has 5 training errors.

The rule assigns the same class to every point in its box. The only number it can attach is the share of fraud among the training rows in the box, the same for A (at the corner) and B (deep inside). C is a transaction unlike any row in the table; the rule classifies it as honest without any indication that the answer is unsupported.

What a Rule Can Not Say

- ✗ **How sure.** One share for the whole box — A and B alike.
- ✗ **Honestly.** The share is measured on the rows that chose the box: optimistic, 100% for a box searched until pure.
- ✗ **Under a new base rate.** The training mix is baked into the thresholds.
- ✗ **That an input is strange.** No model of x: C gets an answer like any other.

Examples

- **Fraud rule engines:** "amount > 1k € and new device → block".
- **Decision trees:** rules learned by splitting, one class per leaf.

A rule is a **discriminant function**: $x \mapsto$ a class. No distribution is modelled.

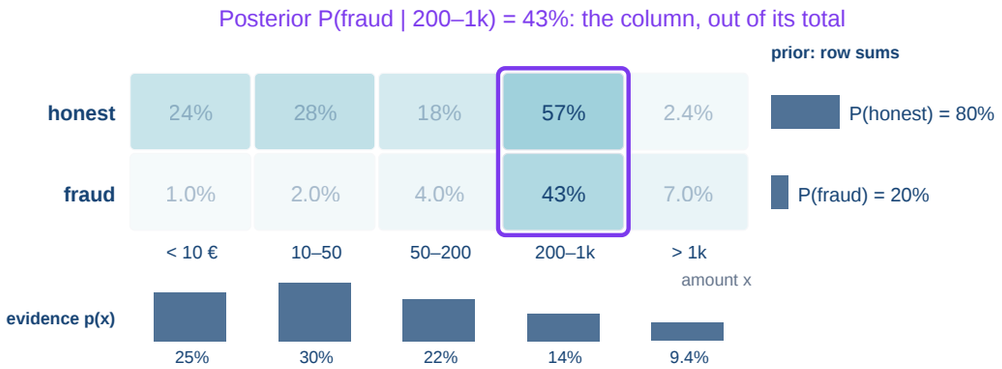
Speaker notes

A discriminant function maps an input directly to a class (Bishop, *Pattern Recognition and Machine Learning*, §1.5.4, distinguishes it from the two probabilistic approaches that follow). Rule sets and decision trees are the standard examples; a support vector machine is another.

The leaf or box share is a relative frequency $\hat{P}(y \mid x \in R)$ for the region R, not $P(y \mid x)$: it is constant over R, and because R was selected to make the share extreme, the training share overstates the share on new data (selection bias; the same effect makes a fully grown decision tree report 100% in every leaf). Calibrating such scores needs held-out data (a held-out set, not the training rows). Thresholds learned on a sample with one base rate encode that base rate; under a different one they are no longer optimal, and nothing in the rule says how to adjust them. Since no distribution over x is represented, the rule cannot measure how typical an input is or handle a missing attribute other than by ad hoc defaults.

The Joint Distribution $p(x_1, \ldots, x_p, y)$

A **probabilistic** classifier starts from the joint distribution of data and hypothesis. Every question is read off it:



question	read off the joint
Is this transaction fraud?	$P(y \mid x) = p(x, y) \,/\, p(x)$
How does fraud look?	$p(x \mid y) = p(x, y) \,/\, P(y)$
How common is fraud?	$P(y) = \sum_x p(x, y)$
How unusual is this transaction?	$p(x) = \sum_y p(x, y)$
The distance is missing — now?	sum x_{dist} out, then condition

> The joint is the course's ideal: model everything, then condition.

Speaker notes

All quantities are obtained by the two operations of Lecture 02: conditioning (divide by a margin) and marginalisation (sum, or integrate, over the variables that are not asked about). For continuous attributes the sums over x are integrals and the probabilities densities.

The heatmap is the joint $p(y, x)$ of the class and one attribute (the amount, five bins), with a fraud share of 20% so that the fraud row is visible. Steps: the cells of the joint; the margins (prior $P(y)$ as row sums, evidence $p(x)$ as column sums); the fraud row against its total, the likelihood $p(x \mid \text{fraud})$; one column against its total, the posterior $P(y \mid x)$. For 200–1k €: $p(\text{fraud}, x) = 0.2 \cdot 0.30 = 0.06$ and $p(\text{honest}, x) = 0.8 \cdot 0.10 = 0.08$, so $P(\text{fraud} \mid x) = 0.06/0.14 \approx 0.43$.

The joint therefore contains strictly more than the decision needs. The two probabilistic classifiers differ in how much of it they model: the discriminative classifier only the column shares, $P(y \mid x)$, as a function of x; the generative classifier the whole picture, factorised as $p(x \mid y)\, P(y)$.

Discriminative Classifier

A **discriminative classifier** models only $P(y \mid x; \theta)$ — nothing about which x occur.

Example: logistic regression, 3 numbers:

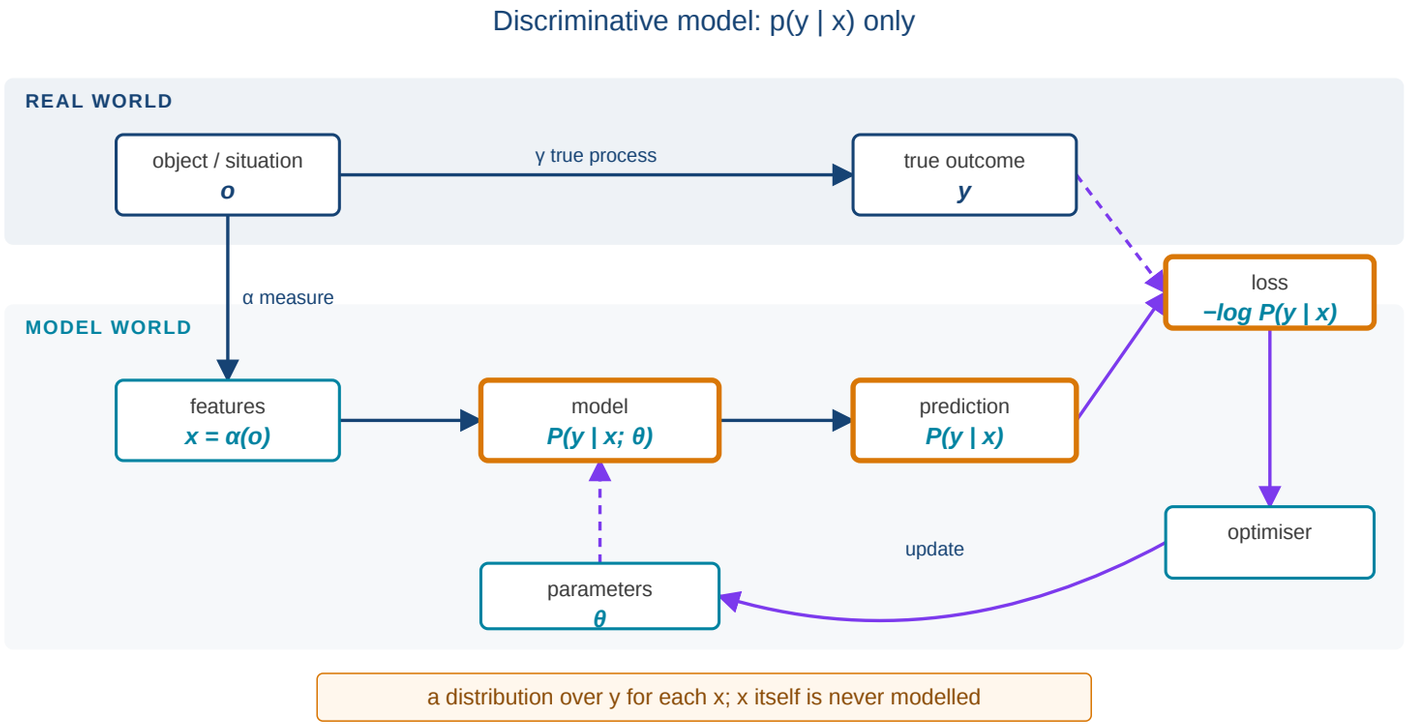
$$z = \theta_0 + \theta_{\text{amt}} \log x_{\text{amt}} + \theta_{\text{dist}} \log x_{\text{dist}}$$

$$P(\text{fraud} \mid x; \theta) = \sigma(z) = 1/(1 + e^{-z})$$

Training: the most probable labels,

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \log P(y_i \mid x_i; \theta)$$

x_i stands only right of the bar: given, never modelled.



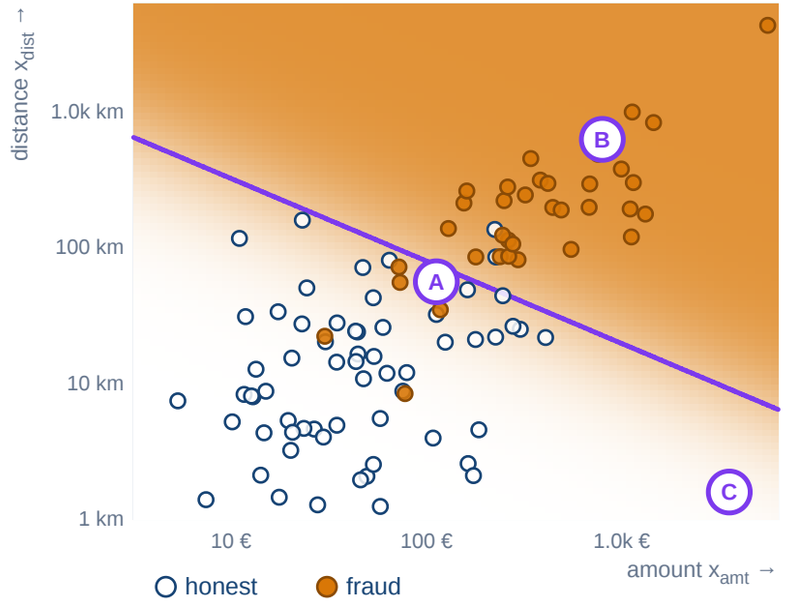
Speaker notes

A discriminative classifier is a family $P(y \mid x; \theta)$ of distributions over the classes, one for every input. It is not a table of the joint read by columns: it is a function of x with a fixed, small number of parameters, which assigns a probability to inputs it has never seen. Logistic regression uses a linear score z of the attributes passed through the logistic function σ, which maps any number to $(0, 1)$; z itself is the log-odds, $z = \log \frac{P(\text{fraud}|x)}{P(\text{honest}|x)}$. A neural network with a softmax output layer uses a learned score. The attributes enter on the log scale, as in the figures.

The learning system of the introductory lecture with a discriminative model in the model box: the prediction is a distribution over the classes, from which a class is chosen; the loss is the negative log-probability of the observed label, $-\log P(y_i \mid x_i; \theta)$, the cross-entropy, and training maximises $\sum_i \log P(y_i \mid x_i; \theta)$, the probability of the training labels given the training attributes (maximum likelihood, made precise in Lecture 04). The features enter only as the condition: the model is evaluated at the observed x and makes no statement about which x occur, so $p(x)$, $p(x \mid y)$ and the prior cannot be recovered and Bayes' theorem cannot be applied to it. A change of base rate must be corrected from outside (Lecture 04).

$P(y \mid x)$ From Data

A and B now differ; C (3.5k € at 1.6 km) still gets an answer, without warning



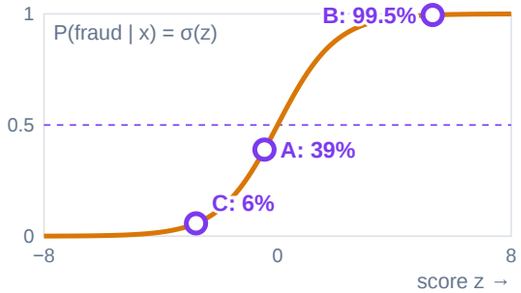
Logistic regression

$z = -11.5 + 2.2 \log x_{amt} + 3.7 \log x_{dist}$

$P(\text{fraud} \mid x) = \sigma(z) = 1 / (1 + e^{-z})$

z: a score, the log-odds — any number

×10 amount: odds ×9 · ×10 distance: odds ×40



7 of 98 training errors · no p(x | y), no p(x)

Compare with the rule: what is new for A and B — and what is still missing for C?

Speaker notes

The 98 transactions of the rule, now with a logistic regression fitted to them by maximum likelihood (Newton's method): $\hat{\theta} = (-11.5,\ 2.2,\ 3.7)$ on $\log_{10}$ amount and $\log_{10}$ distance. The colour is $P(\text{fraud} \mid x; \hat{\theta})$ for every x, the violet line the 0.5 boundary — a straight line, because the score is linear in the attributes. Three numbers replace the box's two thresholds.

The weights are not probabilities. They act on the score $z = -11.5 + 2.2 \log_{10} x_{\text{amt}} + 3.7 \log_{10} x_{\text{dist}}$, the log-odds of fraud, which can be any number; the curve on the right, $\sigma(z)$, turns it into a probability. Read on the odds, a weight is a factor: ten times the amount adds 2.2 to z and multiplies the odds of fraud by $e^{2.2} \approx 9$, ten times the distance by $e^{3.7} \approx 40$. How much that changes the probability depends on where on the curve a transaction sits: near $z = 0$ (A) a step moves the probability a lot, far out (B) hardly at all. A weight's size also depends on the scale of its attribute, so it measures importance only for attributes on comparable scales.

A gets 39%, B 99.5%: the model is graded where the rule was flat (94% for both). C, at 3.5k € and 1.6 km, gets 6% fraud — a confident "honest" for an input unlike any training row, without any warning, because the model has no $p(x)$ to tell it so. The fit has 7 training errors against the rule's 5: logistic regression maximises the probability of the labels, not the number of correct decisions, and its boundary is a straight line where the rule's is a corner.

Generative Classifier

A **generative classifier** models how the data arise — $p(x \mid y; \theta)$ and $P(y; \theta)$ — and obtains $P(y \mid x)$ by Bayes' theorem.

Example: a normal per class and the fraud share, 11 numbers:

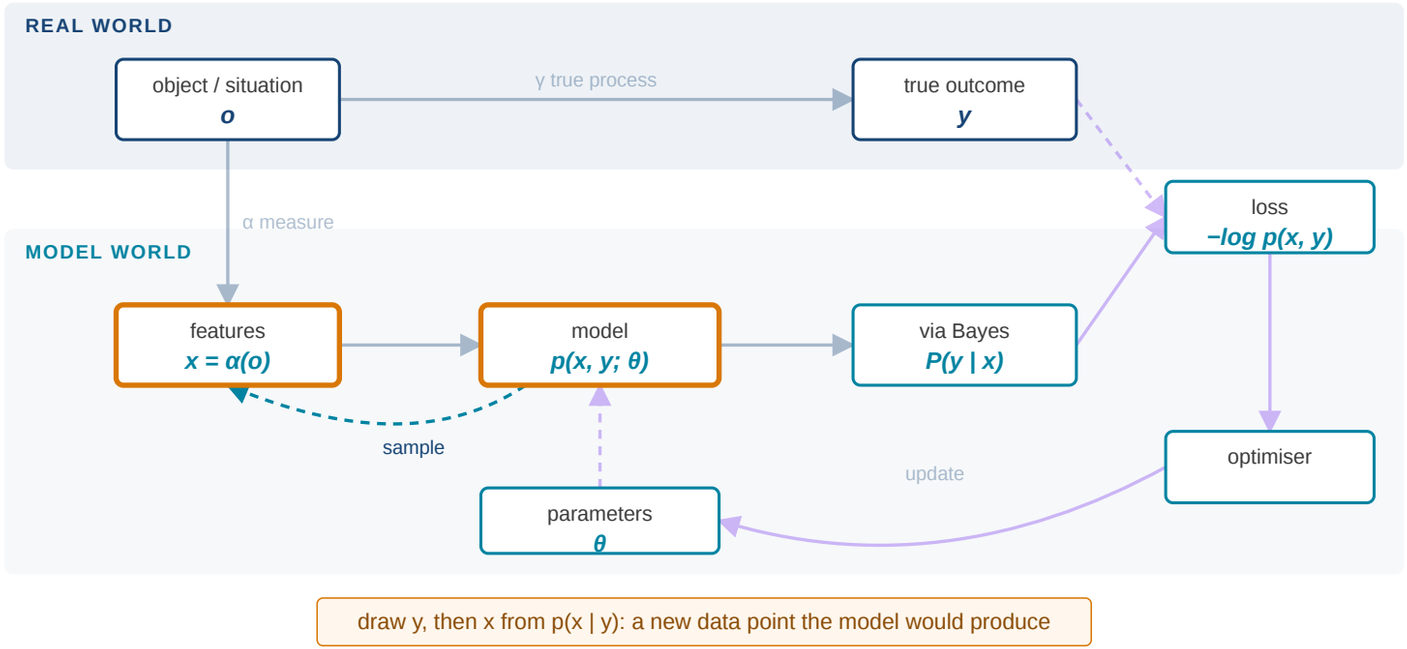
$$p(x \mid y; \theta) = \mathcal{N}\big((\log x_{\text{amt}}, \log x_{\text{dist}});\ \mu_y, \Sigma_y\big)$$
$$P(\text{fraud}; \theta) = \pi$$

Training: the most probable rows,

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \big[\log p(x_i \mid y_i; \theta) + \log P(y_i; \theta)\big]$$

x_i is scored too: modelled, not only given.

Generative model, sampling



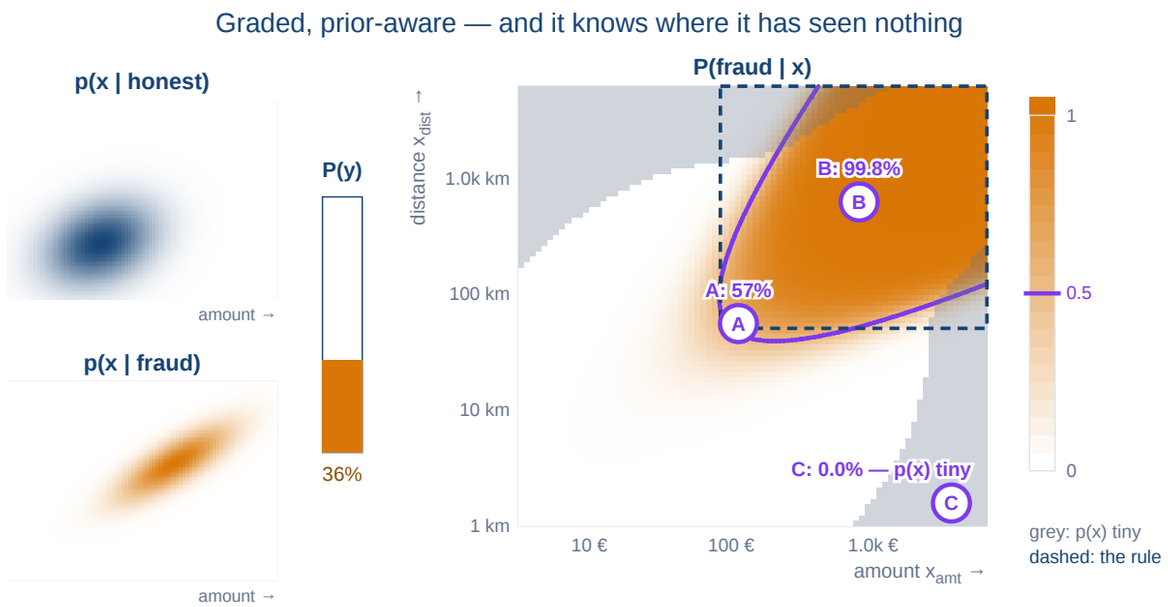
draw y, then x from p(x | y): a new data point the model would produce

Speaker notes

The learning system with a generative model in the model box: the model is the joint $p(x, y; \theta)$; the prediction $P(y \mid x)$ follows by Bayes' theorem; the loss under maximum likelihood is $-\log p(x, y; \theta) = -\log p(x \mid y; \theta) - \log P(y; \theta)$, which scores the features as well as the label, so the features box is part of what is fitted. The second step runs the model without input: from the parameters to a sampled data point.

In the example μ_y is the mean of the class in (log amount, log distance) and Σ_y its covariance matrix — two spreads and a correlation, introduced properly in the module on continuous attributes. The maximisation has a closed form: per class the mean vector, spreads and correlation of its rows, and $\hat{\pi}$ the share of fraud rows, 35 of 98. Parameter count: $2 \cdot (2 + 2 + 1) + 1 = 11$, against 3 for logistic regression and 2 for the rule — the price of modelling x.

$p(x \mid y)$ From Data



Set $P(\text{fraud})$ to 0.1%. What happens to the boundary — and why can the rule not do this?

Speaker notes

Each class has its own distribution over both attributes, a two-dimensional normal in (log amount, log distance) fitted to the class's rows of the same 98 transactions: two means, two spreads and a correlation per class. The prior starts at the training share, 36%. Bayes' theorem combines them for every point, $P(\text{fraud} \mid x) = p(x \mid \text{fraud})P(\text{fraud}) \,/\, \sum_{y'} p(x \mid y')P(y')$, which the large panel shows; the violet line is where the posterior is 0.5.

Step 3 adds the rule (dashed), greys out the region where $p(x)$ is tiny and marks A, B and C: 57%, 99.8% and 0.0% — and C lies in the grey region. There the posterior is still defined, but $p(x)$ says that no training data supports it; this is the warning neither the rule nor logistic regression can give.

Lowering the prior shrinks the fraud region: the likelihoods are unchanged, only the factor $P(\text{fraud})$ moves, so the model can be carried to a setting with a different base rate without refitting. At 1 in 1,000 the boundary retreats to where the likelihood ratio exceeds 999. The sliders move the fraud mean and the correlation within the fraud class; the button draws transactions from the model.

Generating Data

Generating is sampling from the model: draw $y \sim P(y)$, then $x \sim p(x \mid y)$ — here y = spam, x = the words of a mail.



Which mail reads like a real one — and what does naive Bayes leave out?

Speaker notes

Sampling follows the factorisation $p(x, y) = p(x \mid y)P(y)$ from left to right: first the class, then the attributes given the class; the figure fixes the class to spam, and the attributes are the words of the mail. Both models share one vocabulary of 13 words, listed once in the middle of the chart: naive Bayes' probabilities to the left, the language model's to the right, on the same scale. Each step draws one word in both mails (top) and shows the distribution over the vocabulary it was drawn from (bottom). Naive Bayes draws every word from the same $p(\mathrm{word} \mid \mathrm{spam})$: each word is typical of spam, the sequence is not a sentence — the dependences between words are exactly what the naive assumption (next module) drops. The language model draws each word from $p(x_t \mid x_1, \ldots, x_{t-1})$, the chain rule of probability, so the candidates change with the context: after "won" only "a" is possible, after "a" only "free" or "prize". Its bars are peaked where naive Bayes' bars are flat. Both models are toys with hand-set probabilities, and the toy language model conditions only on the previous word; a real one conditions on the whole context with one shared neural network and is fitted by maximum likelihood (cross-entropy on the next token). "Sample again" draws two new mails.

Drawing a missing attribute from $p(x_j \mid x_{-j}, y)$, the conditional given the observed attributes, fills in part of a data point. Neither a rule nor a model of $P(y \mid x)$ can generate: generating needs a model of x. Whether a model can generate is a property of *what* it models, not of *how* its parameters are fitted. Each factor of a language model, taken alone, is a classifier over the vocabulary; the product of all factors is the generative model.

Three Kinds of Classifier

	rule	discriminative	generative
models	$x \mapsto$ class	$P(y \mid x; \theta)$	$p(x \mid y; \theta)\, P(y; \theta)$
fitted here	2 thresholds	3 weights	11 numbers
A, just inside the box	fraud, 94%	39%	57%
B, deep inside	fraud, 94%	99.5%	99.8%
C, unlike any row	honest, **no warning**	6%, **no warning**	0.0%, **p(x) tiny**
new base rate	✗ refit	correct from outside	✓ swap $P(y)$
missing field	✗ ad hoc	✗ ad hoc	✓ sum it out
generate data	✗	✗	✓
examples	decision tree	logistic regression	naive Bayes, language model

> The price of the right column: a distribution over **all** attributes, per class.

Speaker notes

All three were fitted to the same 98 transactions. The rule gives A and B the same box share; logistic regression and the generative model grade them. For C, unlike any training row, the rule and logistic regression answer without any indication that the answer is unsupported; the generative model's answer is just as confident (0.0% fraud) but comes with $p(x)$, which says that no data lie there.

New base rate: the posterior odds are the prior odds times the likelihood ratio, so a generative model only replaces $P(y)$, provided the class-conditional distributions are the same in both settings. A discriminative model absorbs the training base rate into its parameters; it can be corrected by multiplying its odds by the ratio of new to old prior odds, which requires knowing the training base rate. A rule has to be re-learned.

Missing attributes: $p(x_{\text{obs}} \mid y) = \sum_{x_{\text{mis}}} p(x_{\text{obs}}, x_{\text{mis}} \mid y)$. Without a model of x the alternatives are imputation or a separate model per pattern of missing fields.

The comparison of accuracy, which favours the discriminative classifier with much data and the generative one with little, follows in Lecture 04.

The Curse of Dimensionality

Guiding Question: A generative classifier needs $p(x \mid y)$ over all attributes at once. How many numbers is that — and how can a model get by with fewer?

Bayes' Theorem With Two Red Flags

Two flags at once: D = new device, F = foreign IP.

$$P(\text{fraud} \mid D \cap F) = \frac{P(D \cap F \mid \text{fraud}) \cdot P(\text{fraud})}{\sum_y P(D \cap F \mid y)\, P(y)}$$

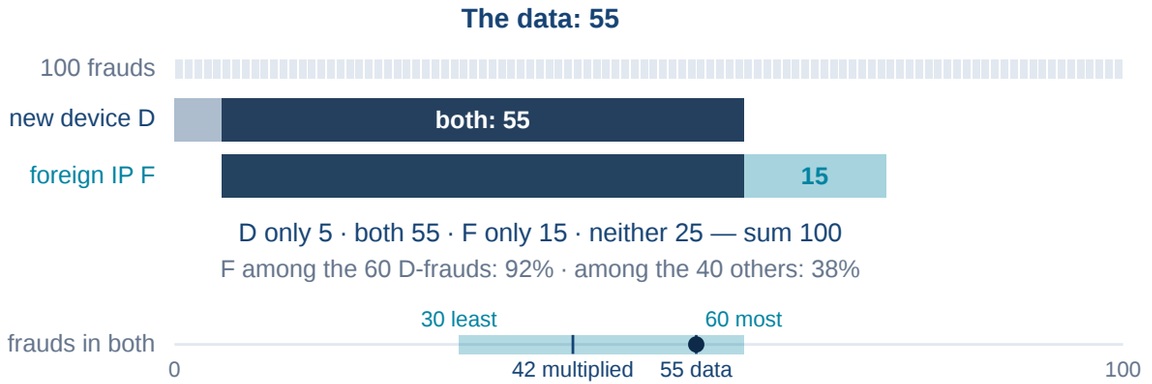
Nothing in the theorem changes. New is the evidence: a **conjunction** $D \cap F$, whose likelihood is needed for every class y.

Among 100 frauds, 60 show D and 70 show F. How many show both?

Speaker notes

This is Bayes' theorem for multiple events of Lecture 02 with the classes as the partition and the evidence $B = D \cap F$. Nothing in the theorem changes; the difficulty is entirely in the likelihood of the conjunction, which is needed for every class: the denominator sums $P(D \cap F \mid y)\, P(y)$ over fraud and honest. (A conjunction on the other side, $P(E_i \cap E_k \mid B)$ for two classes, is zero: classes are disjoint. Several events enter a classifier as several pieces of evidence.)

The Joint of Two Flags

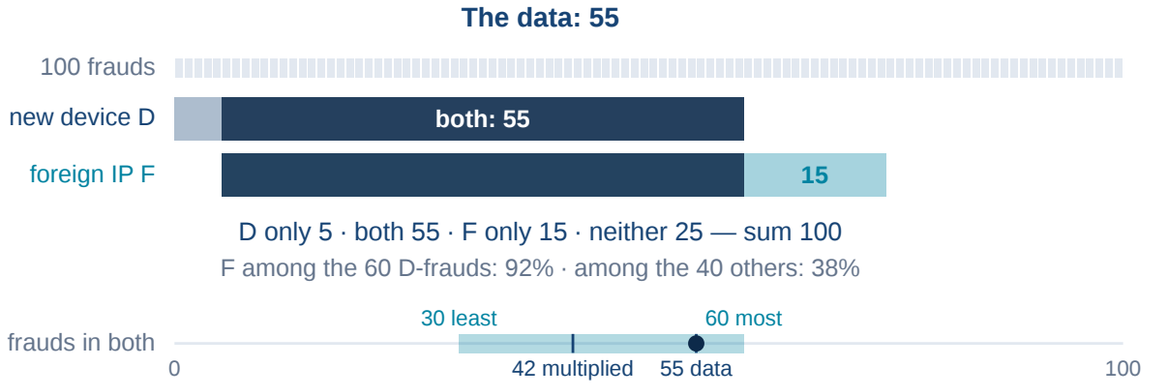


Speaker notes

The bounds, read off the bars: F needs 70 of the 100 frauds and only 40 lack D, so at least 30 frauds show both; and no more than the 60 frauds with D can. In general, for probabilities a and b the intersection lies in $[\max(0, a+b-1), \min(a, b)]$ (Fréchet bounds), here $[0.30, 0.60]$. Every value in between is a different joint with the same two margins; the two rates fix two numbers, the joint of two flags has three.

The positions of F mean different dependences. At 42 the share of F is the same with and without D (70% each): independence given the class, the only case in which the product is right. In the running example 55 frauds show both: 92% of the frauds with a new device come from a foreign IP, but only 38% of the others ($15/40$). An account take-over brings both flags at once, so they are positively dependent among frauds. Among honest transactions the figures are 0.10, 0.05 and 0.04 for both instead of the product 0.005; both values return in Lecture 03.

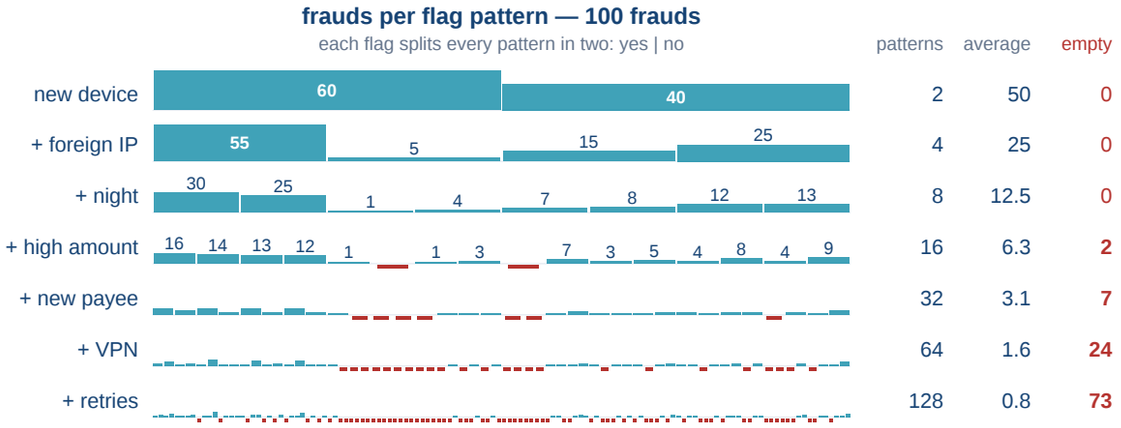
The Joint of Two Flags



Not fixed by the parts: 30 to 60 all fit both rates, so $P(D \cap F \mid y)$ is a number of its own. The product $0.60 \cdot 0.70 = 0.42$ assumes D and F **independent given the class** — naive Bayes.

One Number per Flag Pattern

A third flag, *night*: how many groups now — and how many frauds in each?



Speaker notes

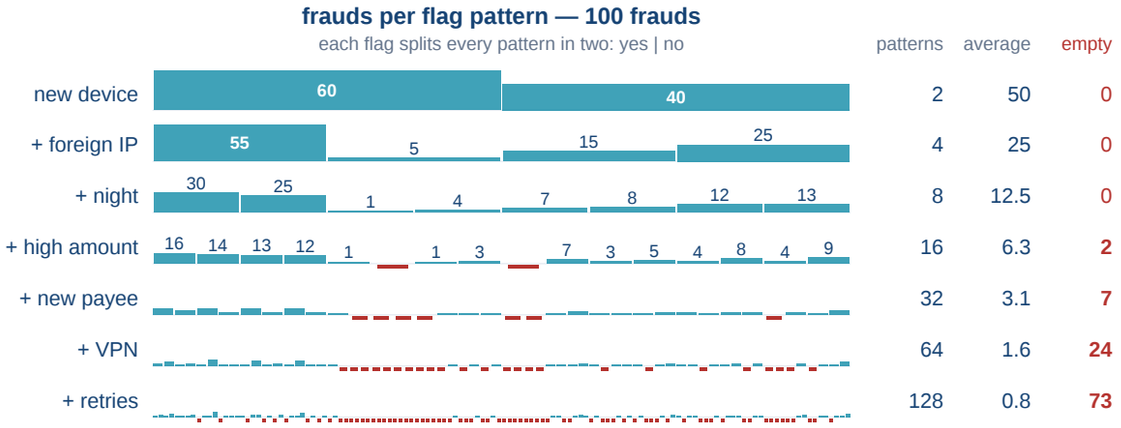
The rows nest: each bar of a row splits into the two bars below it, flag present on the left, absent on the right. Row 2 is the four groups of the previous slide (55, 5, 15, 25). The probabilities of all patterns of a class sum to 1, so one of them is determined by the others: 2^p patterns give $2^p - 1$ free numbers per class, and the two classes need them separately.

Exactly, with the chain rule, $P(E_1 \cap \ldots \cap E_p \mid y) = P(E_1 \mid y)\, P(E_2 \mid E_1, y) \cdots P(E_p \mid E_1, \ldots, E_{p-1}, y)$: each later factor conditions on more flags, so it is estimated only from the rows that match all of them — one row of the figure further down. Dropping the conditioning flags in every factor gives $\prod_j P(E_j \mid y)$, which is exact iff the flags are conditionally independent given the class (naive Bayes).

The flags after D and F are illustrative (drawn with a fixed seed, with night, a new payee and a VPN more frequent in an account take-over). In this sample the first patterns become empty at 4 flags (2 of 16), then 7 of 32, 24 of 64 and 73 of 128. An empty pattern has count 0, so its estimate is 0 — the problem the next slides scale up to 30 flags. On average the frauds per pattern halve with every flag ($100/2^p$); the honest class has a thousand times more rows and therefore, on average, about ten more flags ($2^{10} \approx 1{,}000$) before the same happens.

One Number per Flag Pattern

A third flag, *night*: how many groups now — and how many frauds in each?



Every flag **doubles** the patterns — p flags, 2^p patterns, $2^p - 1$ numbers per class — and **halves** the frauds behind each: at 7 flags, 73 of 128 patterns are **empty**.

The Training Table Is a Sample

amount (€)	country	hour	new device?	… 30 flags	fraud?
49.90	DE	14	no	…	no
1,250.00	BR	3	yes	…	**yes**
12.50	DE	19	no	…	no

Each column is a **random variable** X_j, each row one **draw** from $p(x, y)$. The fraud rows are draws from $p(x \mid \text{fraud})$.

Estimate $p(x \mid \text{fraud})$ by counting: of 100 frauds, how many look exactly like the next transaction?

Speaker notes

This is the reading of a table set up in Lecture 02: a column is a random variable, a row is one realisation of the vector $(X_1, \ldots, X_d, Y)$, and the table is n draws from a joint distribution nobody has written down. Restricting to the rows with $y = \text{fraud}$ is conditioning on the class, so those rows are a sample from $p(x \mid \text{fraud})$.

Estimating a distribution without any assumption about its form means counting: the estimate of $p(x \mid \text{fraud})$ for a particular vector x is the share of fraud rows equal to x in every attribute. A continuous attribute such as the amount makes exact repeats practically impossible; even after binning it, the number of value combinations grows multiplicatively with every column, as the next two slides count.

The Training Table Is a Sample

amount (€)	country	hour	new device?	… 30 flags	fraud?
49.90	DE	14	no	…	no
1,250.00	BR	3	yes	…	**yes**
12.50	DE	19	no	…	no

Each column is a **random variable** X_j, each row one **draw** from $p(x, y)$. The fraud rows are draws from $p(x \mid \text{fraud})$.

Estimate $p(x \mid \text{fraud})$ by counting: of 100 frauds, how many look exactly like the next transaction?

> Almost always **none**. Counting needs examples of every combination of values.

The Joint Explodes

30
yes/no red flags per transaction

→

$2^{30} - 1$
≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone

→

100
frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

Speaker notes

A joint distribution over 30 binary variables has $2^{30} = 1{,}073{,}741{,}824$ cells; since they sum to 1, $2^{30} - 1 = 1{,}073{,}741{,}823$ are free parameters, per class. With 100 fraud cases at most 100 cells have a non-zero count, so at least 99.99999% of the cells have MLE exactly 0, and any fraud pattern not seen in training receives $p(x \mid \text{fraud}) = 0$ and therefore $P(\text{fraud} \mid x) = 0$ (or an undefined $0/0$ if the pattern was not seen among honest transactions either).

A share computed from a tiny count is an extreme value, here multiplied across a billion cells; naive Bayes meets the same zero in a single cell, and Lecture 04 names the estimate behind it. More data does not solve it: filling the table reliably would need far more frauds than cells. The general pattern, and the two ways to reduce the number of parameters, follow on the next two slides.

The Curse of Dimensionality

attributes	value combinations (10 values each)	frauds per combination (100 frauds)
1	10	10
2	100	1
3	1,000	0.1
30	10^{30}	10^{-28}

Every attribute **multiplies** the cells and **divides** the data per cell. Counting needs data in every cell — a full joint over many attributes can never be estimated by counting.

Speaker notes

With p attributes of m values each, the joint has m^p cells, and n observations put on average n/m^p observations into each. Already at three attributes with ten values the average cell holds 0.1 frauds, so most cells are empty and their counts estimate 0. The name "curse of dimensionality" (Bellman) covers this exponential growth of the space with the number of dimensions; it affects every method that relies on having nearby examples, not only counting. Continuous attributes make it worse: without binning every value is its own cell.

The required amount of data grows exponentially in p, while real datasets grow by collecting more rows, not by an exponential factor per added attribute. Adding attributes therefore makes an unrestricted estimate worse, not better, unless the model is restricted.

Two Ways Out

① Simplify the structure

Assumption on the data: attributes **independent given the class —** one small table each, not one huge joint.

30 flags: **30** numbers per class, not $2^{30} - 1$.

NAIVE BAYES — TODAY; GRAPHS — LECTURE 05

② Use a parametric family

Assumption on the model: a **formula with few parameters** instead of a table of counts.

amount given fraud ~ log-normal(μ, σ): **2** numbers, not 100 bins.

LECTURE 02; A NORMAL PER CLASS — TODAY

Both are **assumptions**. They make the model estimable — and wrong wherever the world disagrees with them.

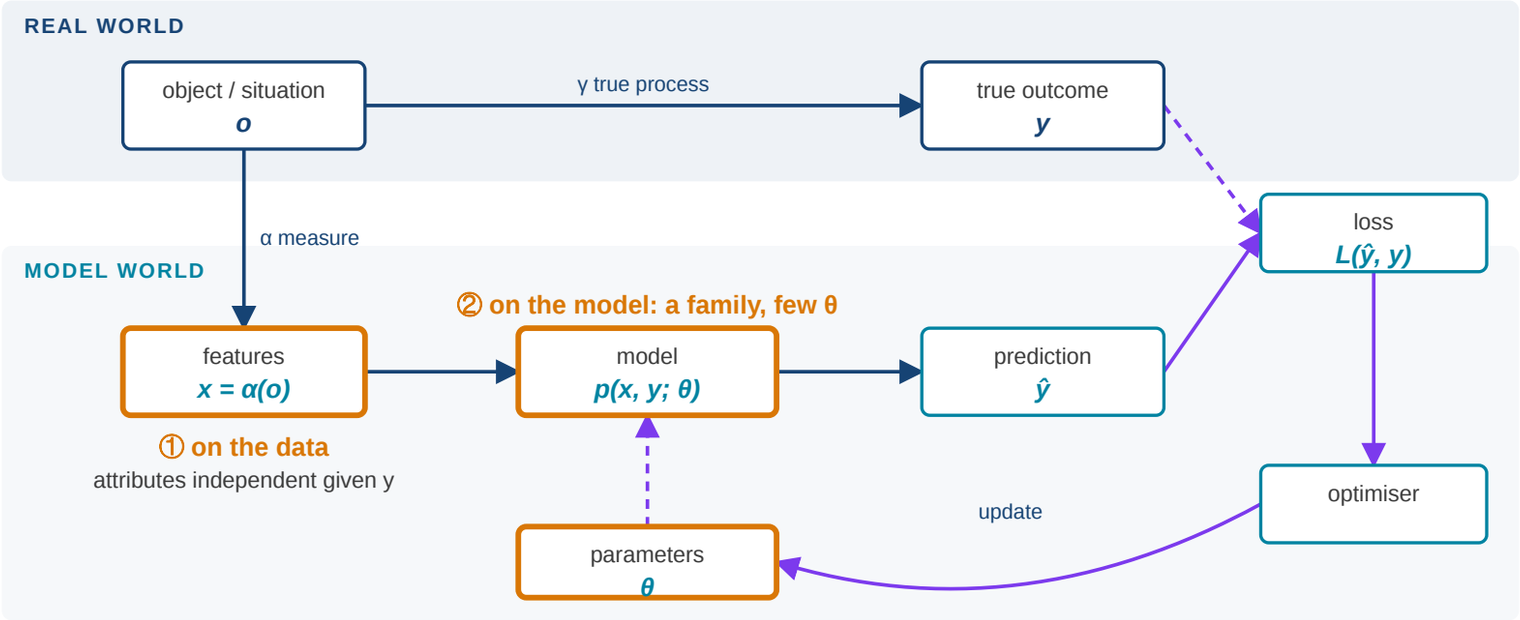
Speaker notes

The two ways reduce the number of parameters at different places. Structure: conditional independence factorises $p(x \mid y) = \prod_j p(x_j \mid y)$, so the parameter count grows linearly in the number of attributes instead of exponentially; a Bayesian network keeps selected dependences and still factorises. Parametric families: each factor is described by a family with a fixed small number of parameters (Bernoulli for a flag, categorical for the country, normal or log-normal for an amount), so a continuous attribute needs no bins at all.

Naive Bayes uses both: independent attributes given the class, each with a small family. The resulting estimates are stable from little data, but correlated attributes are counted twice and a misfitting family distorts the likelihood ratio. The naive Bayes module takes the first cost up; continuous attributes (Lecture 03) the second, and show that a family alone — one Gaussian per class — already turns the exponential count into a quadratic one.

Assumptions on the Data and on the Model

② Use a parametric family: an assumption on the model



which distributions the model may use: a family — it fixes how many θ are fitted

Speaker notes

The figure is the learning system of Lecture 01, in this lecture's notation: the model box computes $p(x, y; \theta)$. Both ways out restrict that joint, but they restrict different things. Simpler structure is a claim about how the data arise: which attributes depend on each other once the class is known. It holds or fails in the world, and the data can show it — among frauds 92% of the new-device cases come from a foreign IP against 38% of the others, so the independence of naive Bayes fails there. A parametric family is a choice of the model: the shape each distribution may take (Bernoulli for a flag, a normal for a log amount) and with it the number of parameters θ. It is wrong when the shape is, for example one normal for a class with two fraud patterns.

The two choices are independent of each other. Naive Bayes with counted tables simplifies the structure only; one full Gaussian per class uses a family only and keeps every dependence; Gaussian naive Bayes does both (Lecture 03). A graphical model (Lecture 05) simplifies the structure less drastically, keeping selected dependences.

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

Speaker notes

Each class needs a full distribution over the $2^3 = 8$ flag patterns, which has $8 - 1 = 7$ free numbers because the probabilities sum to 1. Two classes give 14; the class prior $P(\text{fraud})$ adds one more ($P(\text{honest}) = 1 - P(\text{fraud})$ is determined). Total: 15. Typical miscounts are 16 (counting both class probabilities) and 17 (counting all 8 cells per class).

General formula for K classes and p binary attributes: $K(2^p - 1) + (K - 1)$. Under the naive assumption it becomes $Kp + (K - 1)$, for the example $2 \cdot 3 + 1 = 7$.

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

The generative route models **how the data arises** — $p(y)$ and $p(x \mid y)$ — and converts with Bayes. Counting that joint needs data in every combination of values, which **explodes** with the number of attributes. The way out is an assumption: **simpler structure**, **parametric families**, or both.

Naive Bayes

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

Definition. A and B are **conditionally independent given** C, written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \quad \Longleftrightarrow \quad P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B tells us nothing more about A.**

Among frauds only: does a new device make a foreign IP more likely?

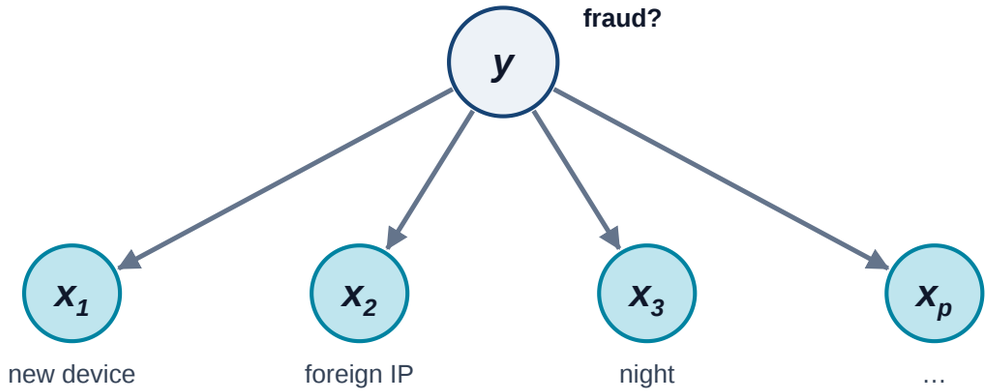
Speaker notes

The two forms are equivalent whenever $P(B \cap C) > 0$: dividing $P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$ by $P(B \mid C)$ gives $P(A \mid B \cap C) = P(A \mid C)$. The relation is symmetric in A and B. For random variables it must hold for every combination of values, and the condition is evaluated separately within each value of C.

Conditional and marginal independence are different properties; neither implies the other. Two flags can be dependent overall because both are more frequent among frauds, yet independent within each class; conversely, conditioning can create a dependence between independent causes (Lecture 05 treats both cases with a counted example).

Answer to the question: in an account take-over the attacker typically uses both a new device and a foreign IP, so among frauds the two flags are positively dependent. With the figures used later in this module, $P(F \mid D, \text{fraud}) = 0.55/0.60 \approx 0.92$, compared with $P(F \mid \text{fraud}) = 0.70$. This is the dependence that naive Bayes assumes away.

The Naive Assumption



$p(y) \cdot p(x_1 \mid y) \cdot p(x_2 \mid y) \cdot \ldots \cdot p(x_p \mid y)$

Assume the attributes are **independent given the class**:

$$p(x_1, \ldots, x_p \mid y) = \prod_{j=1}^{p} p(x_j \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

Shortcut 1 — assume it apart. Dropped: every dependence *within* a class.

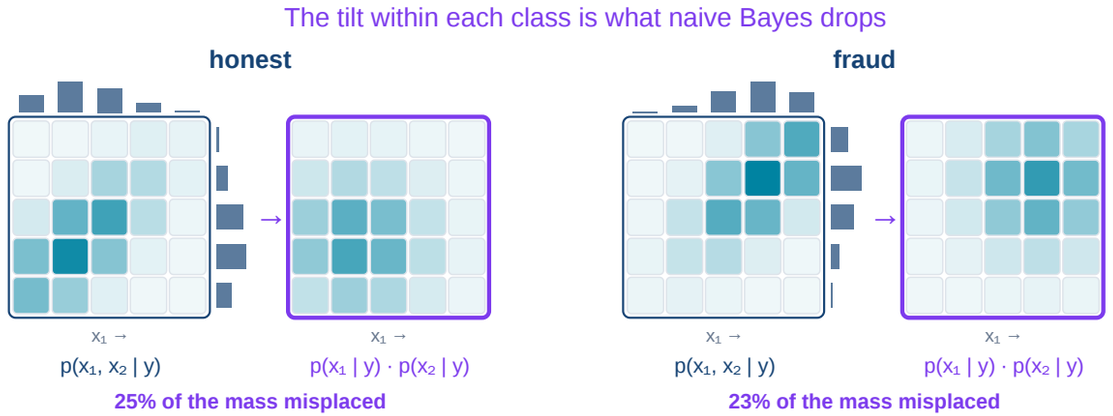
Speaker notes

In the graph the class y points to every attribute and no arrow joins two attributes. Read as a Bayesian network (Lecture 05), each missing arrow is an independence claim: every attribute is independent of all others given the class. The attributes are still dependent marginally, through the class.

Parameter count: with the assumption, $p(x \mid y)$ needs one Bernoulli parameter per binary flag and class, p numbers instead of $2^p - 1$. For a categorical attribute with k_j values the count is $k_j - 1$ per class. The three-flag example drops from 15 numbers to $2 \cdot 3 + 1 = 7$; for 30 flags from about $2.1 \cdot 10^9$ to 61.

The model is still generative, since it specifies $p(y)$ and $p(x \mid y)$ and can generate transactions. What it drops is every dependence between attributes within a class (shortcut 1 of the course roadmap). The assumption is almost never exactly true; whether it matters depends on the task, as the last slides of the module show.

Assuming It Apart, Drawn



Predict: set the correlation to 0. What happens to the two pictures?

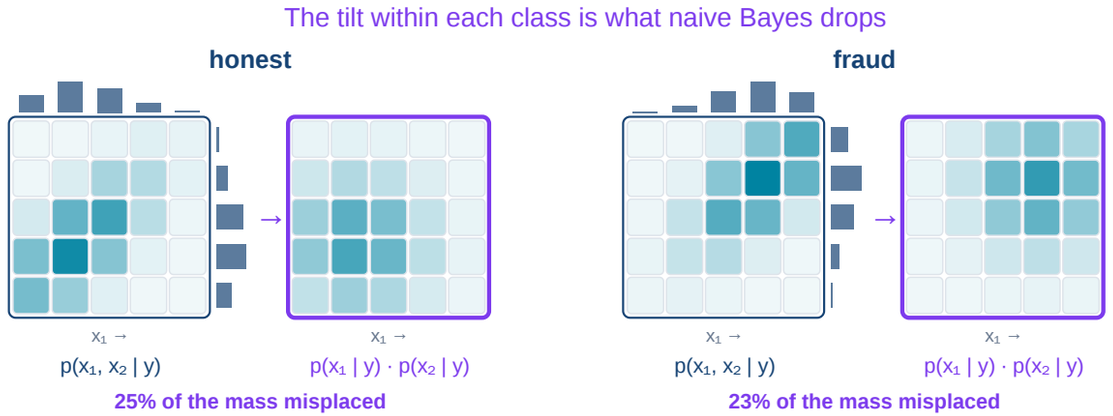
Speaker notes

The full object is a cube, class × x_1 × x_2; the figure shows its two class slices. Left: the true class-conditional joint $p(x_1, x_2 \mid y)$ on a 5×5 grid, $5^2 - 1 = 24$ numbers per class, and $5^p - 1$ for p attributes. Its two margins, $p(x_1 \mid y)$ and $p(x_2 \mid y)$, need $2 \cdot 4 = 8$. Right: the product of the margins, which is what naive Bayes stores. A product of margins is always axis-aligned — the same picture as the independence rectangle of Lecture 02 — so it cannot represent a tilt. The share of probability mass placed differently from the true joint (the total variation distance) is shown per class.

Two dependences must be kept apart. Within a class, a tilted blob means x_1 and x_2 are correlated given y; this is what naive Bayes drops. Across classes, the two blobs sit in different corners, so over all transactions x_1 and x_2 are dependent even when $\rho = 0$ — through the class, which naive Bayes keeps. Conditional independence given y does not imply independence overall.

What naive Bayes stores are the margins $p(x_j \mid y)$: one-dimensional distributions, one per attribute and class. How each is estimated is the next question.

Assuming It Apart, Drawn



Predict: set the correlation to 0. What happens to the two pictures?

They become **identical**. Naive Bayes is exact exactly when the attributes are independent **within** each class.

A Distribution per Attribute and Class

Naive Bayes needs $p(x_j \mid \text{fraud})$ and $p(x_j \mid \text{honest})$ for each attribute on its own. How would you estimate them for these four?

Speaker notes

The naive assumption leaves p one-dimensional distributions per class, and for each the scale of the attribute suggests a model, as in "The Scale Suggests the Model" of Lecture 02. For a nominal attribute the table is the family: a Bernoulli for a flag, a categorical for a country, both fitted by relative frequencies. Naive Bayes over flags and categories is therefore fitted by counting alone, the case taken up next.

A count attribute can be treated either way. As a table, every observed value gets its share: many parameters, and a value never seen among frauds (twelve logins) gets probability 0 — the problem of "One Zero Vetoes Everything" below. As a Poisson, one rate per class describes all values, and its maximum likelihood estimate is the mean count of the class; twelve logins keep a small positive probability. The price is the shape: a Poisson has variance equal to its mean, which real counts often exceed.

A continuous attribute has no table at all without bins. A normal per class on the log amount gives Gaussian naive Bayes, the subject of Lecture 03. The factors may be mixed freely, Bernoulli for some attributes, Poisson or normal for others, because under the naive assumption each factor is estimated on its own. Text classifiers use the same idea with word counts (multinomial naive Bayes). All of these maximum likelihood estimates are closed-form summaries of the class's rows — shares, means, spreads — so no optimiser is needed.

A Distribution per Attribute and Class

Naive Bayes needs $p(x_j \mid \text{fraud})$ and $p(x_j \mid \text{honest})$ for each attribute on its own. How would you estimate them for these four?

attribute x_j	values	model for $p(x_j \mid y)$	fitted from the class's rows by	numbers per class
new device	yes / no	Bernoulli(θ_y)	the share of *yes*	1
country	k countries	categorical	the share of each country	$k - 1$
logins, last hour	0, 1, 2, …	a **table** of the values — or **Poisson** (λ_y)	the share of each value — or the **mean** count	one per value — or 1
amount	positive, skewed	**normal** on the log amount	mean and spread of $\log x$	2

A Distribution per Attribute and Class

Naive Bayes needs $p(x_j \mid \text{fraud})$ and $p(x_j \mid \text{honest})$ for each attribute on its own. How would you estimate them for these four?

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country	k countries	categorical	the share of each country	$k - 1$
logins, last hour	0, 1, 2, …	a **table** of the values — or **Poisson** (λ_y)	the share of each value — or the **mean** count	one per value — or 1
amount	positive, skewed	**normal** on the log amount	mean and spread of $\log x$	2

Naive Bayes fixes the **structure** (way out ①). Each factor still needs a model: a **table**, fitted by counting, or a **family** with few parameters (way out ②). Both are fitted per class, in one pass over the rows.

Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \qquad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y)}{n(y)}$$

One small table per attribute

For flags and categories, each factor is a **table**: a categorical — for a yes/no flag, a Bernoulli.

Estimated by its share

$P(\text{new device} \mid \text{fraud})$: of the 100 frauds, how many came from a new device? 60, so 0.6.

Naive Bayes over tables is fitted in **one pass**, by counting — no optimiser. Why a share is the right estimate — and when it is not — is Lecture 04.

Speaker notes

Derivation. Under the naive assumption the log-likelihood of a labelled table is $\sum_i \log P(y_i) + \sum_i \sum_j \log p(x_{ij} \mid y_i)$. Grouping terms by attribute j and class y gives one term $\sum_v n(x_j = v, y) \log \theta_{jvy}$ per attribute and class, plus one term for the class prior. No parameter appears in two terms, so each is maximised separately, and each is the log-likelihood of a categorical distribution whose maximum likelihood estimate is the relative frequency (derived for two values as h/n in Lecture 04; for k values with a Lagrange multiplier). The categorical distribution was introduced in Lecture 02 as the distribution for a nominal attribute.

The independence assumption is what makes the MLE a closed-form count: with a full joint the parameters are per combination of values, and the counts would be of entire rows. Logistic regression, by contrast, has no closed-form MLE and is fitted iteratively. Every estimated number in naive Bayes is a peak of a likelihood, so shortcut 2, keep the peak, is taken in addition to shortcut 1; Lecture 04 makes this precise.

The Classifier

$$P(y \mid x_1, \ldots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^{p} \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio } \mathrm{LR}_j}$$

> Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

Speaker notes

The first line is Bayes' theorem with the naive product in place of $p(x \mid y)$; the denominator is the law of total probability. In the ratio of two posteriors the denominator cancels, which gives the odds form: posterior odds = prior odds × product of likelihood ratios. Odds and their relation to probabilities, $P = \text{odds}/(1 + \text{odds})$, were introduced with Bayes' theorem in Lecture 02.

Each attribute contributes one factor LR_j. A ratio above 1 raises the odds of fraud, below 1 lowers them, and exactly 1 means the attribute is uninformative. Because multiplication is commutative, the order in which evidence arrives does not matter. An observed absent flag also contributes a factor, $(1 - p_{j,\text{fraud}})/(1 - p_{j,\text{honest}})$, which is usually below 1 for a red flag. For more than two classes, odds relative to one reference class play the same role.

Likelihood Ratios From Counts

The training table: 100,000 transactions, **100** fraud and **99,900** honest.

A new device shows up in 60 frauds — and in 9,990 honest transactions. How strong is it as evidence for fraud?

Speaker notes

Each share is the fitted factor of "Fitting Is Counting": $\hat{P}(x_j = \text{yes} \mid y) = n(x_j = \text{yes}, y)/n(y)$, counted within the class. The likelihood ratio divides the two shares of one flag, $\text{LR}_j = P(x_j \mid \text{fraud})/P(x_j \mid \text{honest})$, and says how many times more common the observation is among frauds than among honest transactions. Arithmetic: $60/100 = 0.60$ and $9{,}990/99{,}900 = 0.10$, so $\text{LR} = 6$; $0.70/0.05 = 14$; $0.50/0.20 = 2.5$.

The ratio measures the evidence, not the answer. Of the $60 + 9{,}990 = 10{,}050$ transactions with a new device only 60 are fraud, so $P(\text{fraud} \mid \text{new device}) = 60/10{,}050 \approx 0.6\%$: the evidence multiplies the odds by 6, from $1:999$ to $6:999$, and the rare class stays rare. A flag observed absent has its own ratio, for the new device $(1 - 0.60)/(1 - 0.10) \approx 0.44$, which speaks against fraud.

Likelihood Ratios From Counts

The training table: 100,000 transactions, **100** fraud and **99,900** honest.

A new device shows up in 60 frauds — and in 9,990 honest transactions. How strong is it as evidence for fraud?

flag	frauds with it	$P(\text{flag} \mid \text{fraud})$	honest with it	$P(\text{flag} \mid \text{honest})$	likelihood ratio
new device	60 of 100	0.60	9,990 of 99,900	0.10	0.60 / 0.10 = **6**
foreign IP	70 of 100	0.70	4,995 of 99,900	0.05	0.70 / 0.05 = **14**
night	50 of 100	0.50	19,980 of 99,900	0.20	0.50 / 0.20 = **2.5**

Likelihood Ratios From Counts

The training table: 100,000 transactions, **100** fraud and **99,900** honest.

A new device shows up in 60 frauds — and in 9,990 honest transactions. How strong is it as evidence for fraud?

flag	frauds with it	$P(\text{flag} \mid \text{fraud})$	honest with it	$P(\text{flag} \mid \text{honest})$	likelihood ratio
new device	60 of 100	0.60	9,990 of 99,900	0.10	0.60 / 0.10 = **6**
foreign IP	70 of 100	0.70	4,995 of 99,900	0.05	0.70 / 0.05 = **14**
night	50 of 100	0.50	19,980 of 99,900	0.20	0.50 / 0.20 = **2.5**

> Strength is a ratio of **shares**, not of counts: a new device is **6 times as common** among frauds — although 9,990 of its 10,050 transactions are honest.

Three Red Flags

Prior odds $100 : 99{,}900 = 1 : 999$. Likelihood ratios: new device **6**, foreign IP **14**, night **2.5**.

A transaction shows all three flags. Is it more likely fraud than not?

Speaker notes

The odds form of the classifier: prior odds times one likelihood ratio per observed flag. The product of the ratios is 210, and the posterior odds are $210 : 999$, so $P(\text{fraud} \mid x) = 210/(210 + 999) = 0.174$.

The answer is no: the transaction is still more likely honest. A combined likelihood ratio of 210 is strong evidence, but the prior odds of 1 : 999 are stronger; the posterior exceeds 50% only when the combined ratio exceeds 999. This is the base-rate effect of Lecture 02. Intuitive judgements that call such a transaction "clearly fraud" neglect the base rate.

The model in this example contains exactly these three flags, all observed present, so only the three ratios enter. In a model with more flags, every flag observed absent contributes its own factor; for night that factor would be $(1 - 0.5)/(1 - 0.2) = 0.625$.

Three Red Flags

Prior odds $100 : 99{,}900 = 1 : 999$. Likelihood ratios: new device **6**, foreign IP **14**, night **2.5**.

A transaction shows all three flags. Is it more likely fraud than not?

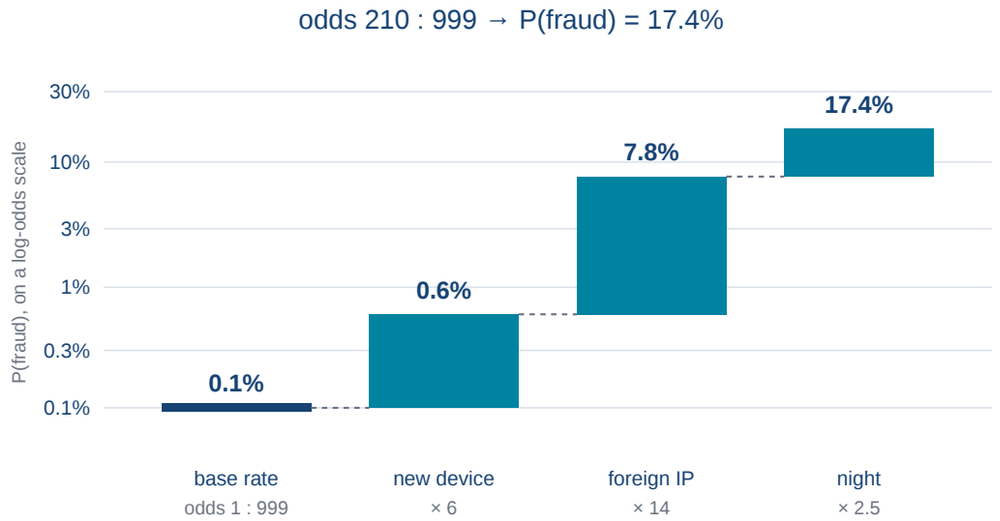
$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \quad \Longrightarrow \quad P(\text{fraud} \mid x) = \frac{210}{1{,}209} \approx 17\%$$

> Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up

Take the logarithm of the odds form: the **product** of ratios becomes a **sum**.

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = \log \frac{P(\text{fraud})}{P(\text{honest})} + \sum_j \log \text{LR}_j$$



Predict before each click: which flag moves the probability most?

Speaker notes

The table is the equation term by term, in natural logarithms: the rounded terms add to $-6.91 + 1.79 + 2.64 + 0.92 = -1.56$, and $P = 1/(1 + e^{1.56}) = 0.174$. The base of the logarithm only rescales the axis; the figure labels it in probabilities.

On a log-odds axis each likelihood ratio is a step of fixed height $\log \mathrm{LR}_j$, independent of the current position: the log-odds are the sum of the log prior odds and the log likelihood ratios.

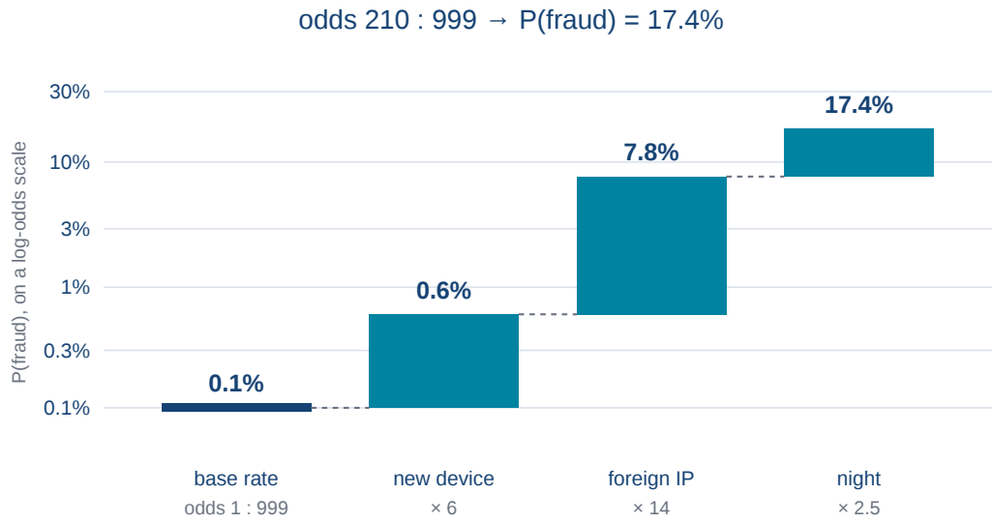
Step 1 (new device): odds $6/999$, probability 0.6%. Step 2 (foreign IP): odds $84/999$, probability 7.8%. Step 3 (night): odds $210/999$, probability 17.4%. The foreign IP has the largest ratio and therefore the largest step; night is the weakest of the three, although it moves the probability more in absolute terms than the first flag did, because probability is not linear in log-odds.

A sum of per-attribute contributions plus a constant is a linear score. For binary attributes the naive Bayes log-odds are exactly linear in x, the same functional form as logistic regression (Lecture 04); the two differ in how the weights are estimated (Lecture 04).

Evidence Adds Up

Take the logarithm of the odds form: the **product** of ratios becomes a **sum**.

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = \log \frac{P(\text{fraud})}{P(\text{honest})} + \sum_j \log \text{LR}_j$$



Predict before each click: which flag moves the probability most?

evidence	odds	log-odds	P
base rate	$1:999$	-6.91	0.1%
new device	$\times 6$	$+1.79$	0.6%
foreign IP	$\times 14$	$+2.64$	7.8%
night	$\times 2.5$	$+0.92$	17.4%
sum	$210:999$	$\mathbf{-1.56}$	**17.4%**

One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

COUNTING, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — a share of a small count, read as certainty.

Speaker notes

This is the zero-frequency problem. The MLE of a count of 0 is the probability 0, and a single factor of 0 makes the whole product 0, so the posterior odds of fraud are 0 regardless of all other likelihood ratios. If a value has count 0 in both classes, the posterior is the undefined $0/0$.

The cause is that a relative frequency of a small count is an extreme value; Lecture 04 shows the same failure on a coin (three heads in three flips give $\hat{\theta} = 1$) and names the estimate. With 100 frauds, any flag with a true rate below about 1% is likely to have count 0 (for a true rate of 1% the probability of no occurrence in 100 is $0.99^{100} = 0.37$). In a log-odds implementation the zero appears as $\log 0 = -\infty$. Standard repairs add pseudo-counts (next slide) or replace zeros by a small floor value; the first has a probabilistic justification.

Laplace Smoothing

Laplace smoothing pretends each of the k_j values was seen **once more**:

$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$.

count among 100 frauds	share	smoothed
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

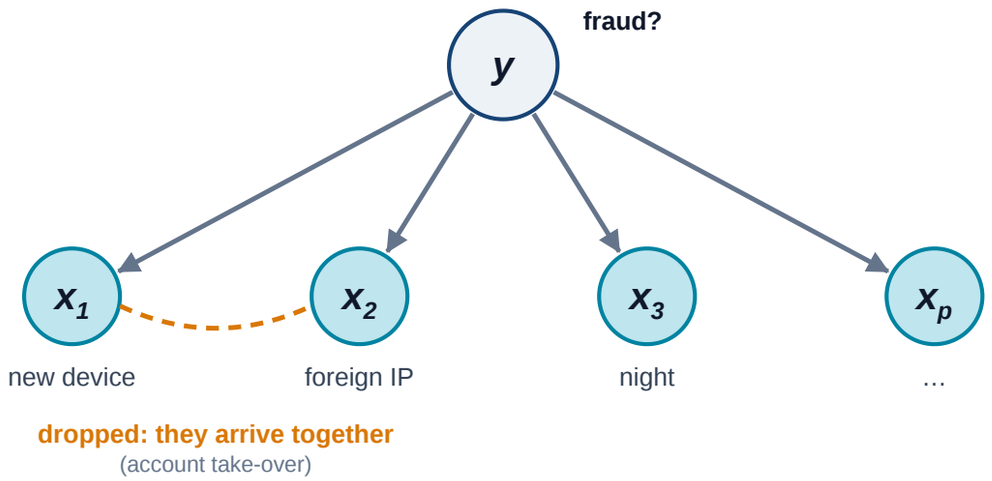
> Where the count is large, nothing changes; where it is 0, the veto is gone. Why **+1**, and not +5? Lecture 04: smoothing is a **prior**.

Speaker notes

The pseudo-counts are added to every value of the attribute, so the smoothed estimates still sum to 1 over the k_j values: $n(y) + k_j$ in the denominator balances one extra count per value. On well-populated cells the change is negligible (0.60 vs 0.598); on empty cells it replaces 0 by $1/102 \approx 0.0098$, so no single factor can force the product to 0.

Add-α smoothing, $(h + \alpha)/(n + 2\alpha)$ for a flag, generalises the rule to α imagined occurrences per value; larger α pulls every estimate more strongly towards one half. The choice $\alpha = 1$ is a convention (Laplace's rule of succession). Lecture 04 derives it: $(h + 1)/(n + 2)$ is the posterior mean of the flag probability under a flat Beta$(1, 1)$ prior (Lecture 04), and add-α is the posterior mean under Beta(α, α); for $k_j > 2$ values the flat Dirichlet plays the same role.

Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ with night
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Speaker notes

The dashed edge in the figure is the dependence between new device and foreign IP that naive Bayes omits. Naive Bayes multiplies both likelihood ratios as if they were independent pieces of evidence, $6 \cdot 14 = 84$. The pair's actual likelihood ratio is $P(D, F \mid \text{fraud}) / P(D, F \mid \text{honest}) = 0.55 / 0.04 = 13.75$. With night added, the posterior odds are $13.75 \cdot 2.5 / 999 = 34.4 / 999$, so $P(\text{fraud} \mid x) = 3.3\%$ instead of 17.4%. Under independence within the classes the pair would occur on $0.6 \cdot 0.7 = 42\%$ of frauds and $0.1 \cdot 0.05 = 0.5\%$ of honest transactions; the assumed 4% among honest ones is eight times that, which is where the error comes from.

Correlated attributes therefore push naive Bayes posteriors towards 0 and 1: the probabilities are not calibrated (Lecture 04). Classification by arg max depends only on which side of the decision threshold the posterior falls, so it can remain correct even when the probabilities are badly off. Domingos and Pazzani (1997, Machine Learning 29) show that naive Bayes can be optimal under zero–one loss in many cases where the independence assumption is violated, which is why it classifies well despite a false assumption.

The word salad of Lecture 03 (Generating Data) is the same assumption seen from the generating side: words drawn independently given the class are each typical of spam, but the dependences that make a sentence are gone — the same dependences whose absence counts correlated evidence twice.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Speaker notes

Answer: up. The duplicated flag contributes its likelihood ratio twice, so the odds become $210 \cdot 14/999 = 2940/999$ and $P(\text{fraud} \mid x) = 2940/3939 = 0.746$, from 17%.

The true information content is unchanged, since a copy of an attribute carries no new evidence: the correct likelihood ratio of the pair (flag, copy) is just the ratio of the flag, 14. Naive Bayes cannot detect duplication, because the model has no parameter for the relation between attributes. A duplicate is the limiting case of perfect correlation; strongly correlated attributes act as partial copies. Real feature sets contain them routinely: the same information coded twice, features derived from others, and indicators that are near-functions of each other. Removing redundant attributes, or a model that captures the dependence (a Bayesian network, Lecture 05), avoids the double counting.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Up — from 17% to about 75%. The odds get a second factor 14: $2{,}940 : 999$. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

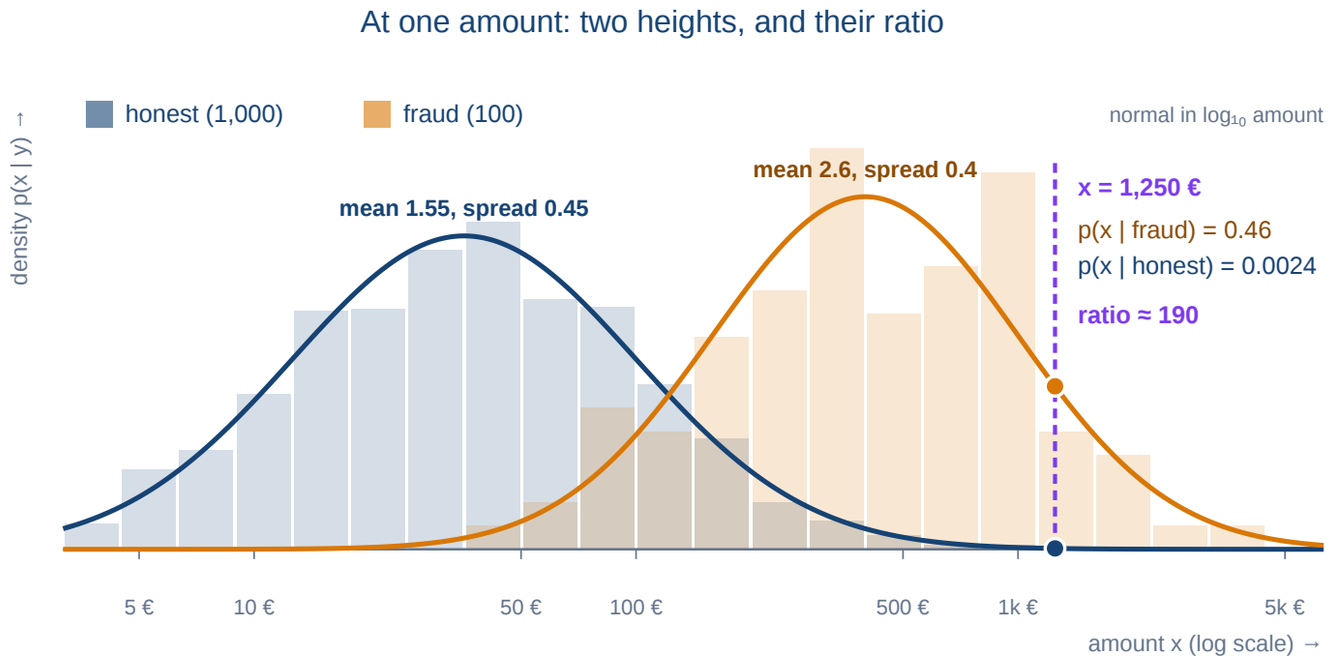
Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and fits every factor by **counting**, smoothed by pseudo-counts so no zero can veto. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.

Continuous Attributes

Guiding Question: An amount of 1,250 € has probability zero. How does a number enter Bayes' theorem — and what does a model of many numbers cost?

A Density per Class



What is the probability that a fraud is exactly 1,250.00 €?

Speaker notes

The histograms are the amounts of the training table, 100 frauds and 1,000 honest transactions (a sample of the honest ones), each class normalised to a density so that both are visible. On a logarithmic axis the amounts of each class are close to normal: the amount itself is log-normal, the second way out of Lecture 03. Step 1 fits one normal per class to $\log_{10}$ amount, with the class's mean and standard deviation as parameters: honest $\hat{\mu} = 1.55$ (median 35 €), $\hat{\sigma} = 0.45$; fraud $\hat{\mu} = 2.6$ (median 400 €), $\hat{\sigma} = 0.40$. Two numbers replace a histogram's bins.

Answer to the question: zero. The amount is a continuous random variable, and every single value has probability 0 (Lecture 02); the curve is a density, probability per unit of x. What Bayes' theorem needs is the ratio of the two densities at the observed value (step 2): at 1,250 € the fraud curve is about 190 times higher than the honest one.

Bayes' Theorem With Densities

For a continuous attribute, $p(x \mid y; \theta)$ is a **density** — probability per unit of x. Bayes' theorem holds with densities:

$$P(y \mid x) = \frac{p(x \mid y; \theta)\, P(y)}{\sum_{y'} p(x \mid y'; \theta)\, P(y')}$$

Why it works. An interval $[x, x + \Delta]$ has probability $\approx p(x \mid y)\, \Delta$ in each class; the Δ cancels.

Normal per class, $\theta_y = (\mu_y, \sigma_y)$:

$$p(x \mid y; \theta) = \frac{1}{\sigma_y \sqrt{2\pi}}\, e^{-(x-\mu_y)^2 / 2\sigma_y^2}$$

Speaker notes

The limit argument: $P(x \le X \le x + \Delta \mid y) = p(x \mid y)\,\Delta + o(\Delta)$, so $P(y \mid x \le X \le x + \Delta) \to p(x \mid y)P(y) / \sum_{y'} p(x \mid y')P(y')$ as $\Delta \to 0$. The conditional on the event $X = x$, which has probability zero, is defined by this limit. The likelihood ratio of a continuous attribute is therefore a ratio of densities, and it depends on the observed value.

Numbers: $x = \log_{10} 1{,}250 = 3.10$. Fraud: $z = (3.10 - 2.6)/0.40 = 1.24$, density 0.46. Honest: $z = (3.10 - 1.55)/0.45 = 3.44$, density 0.0024. Ratio 191, posterior odds $191 : 999$, $P = 0.16$. The densities are taken over $\log_{10}$ amount; over the amount itself both carry the same factor $1/(x \ln 10)$, which cancels in the ratio, so the scale on which the density is written does not change the posterior.

The class's sample mean and standard deviation are the natural estimates (Lecture 02); they are the maximum likelihood estimates of μ_y and σ_y (with divisor n for σ, a difference that vanishes for 100 rows), as Lecture 04 derives for a single parameter.

Bayes' Theorem With Densities

For a continuous attribute, $p(x \mid y; \theta)$ is a **density** — probability per unit of x. Bayes' theorem holds with densities:

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$$p(x \mid y; \theta) = \frac{1}{\sigma_y \sqrt{2\pi}}\, e^{-(x-\mu_y)^2 / 2\sigma_y^2}$$

At 1,250 €: likelihood ratio $0.46 / 0.0024 \approx 190$. With prior odds $1 : 999$, $P(\text{fraud} \mid x) = 190 / 1{,}189 \approx 16\%$.

Gaussian Naive Bayes

Naive Bayes with a **family per attribute** — a normal for each number, a Bernoulli for each flag — all independent given the class:

$$p(x \mid y;\theta) = \prod_{\text{numbers } j} \mathcal{N}(x_j;\ \mu_{jy}, \sigma^2_{jy}) \cdot \prod_{\text{flags } j} \theta_{jy}^{x_j}(1-\theta_{jy})^{1-x_j}$$

attribute of the transaction	fitted per class	likelihood ratio
amount 1,250 €	mean and spread of log amount	≈ 190
new device: yes	share of rows with the flag	6
both		$190 \cdot 6 = 1{,}140$

Speaker notes

The naive assumption of Lecture 03 does not depend on the type of the attributes: each factor $p(x_j \mid y)$ can come from its own family, normal (or log-normal) for amounts and durations, Bernoulli for flags, categorical for the country. The classifier and its odds form are unchanged; every attribute contributes its own likelihood ratio, for a number the ratio of two normal densities at the observed value. The model with only normal factors is called Gaussian naive Bayes (scikit-learn `GaussianNB`).

Parameter count: two numbers per continuous attribute and class (mean and variance), one per binary flag and class. In log-odds form a normal factor contributes $\log(\sigma_{jh}/\sigma_{jf}) - (x_j - \mu_{jf})^2/2\sigma_{jf}^2 + (x_j - \mu_{jh})^2/2\sigma_{jh}^2$, a quadratic function of x_j; with equal variances in both classes the quadratic terms cancel and the contribution is linear in x_j. With unequal variances the ratio eventually falls again on the side of the narrower class: for the amount, $\sigma_{\text{fraud}} < \sigma_{\text{honest}}$, the likelihood ratio peaks at about 3.5 million € and decreases beyond — far outside the data, where $p(x)$ is negligible anyway.

Numbers: $191 \cdot 6 = 1{,}149$; posterior odds $1{,}149 : 999$, $P = 1{,}149/2{,}148 = 0.53$. The amount alone gave 16%, the device flag alone $6/1{,}005 = 0.6\%$.

Gaussian Naive Bayes

Naive Bayes with a **family per attribute** — a normal for each number, a Bernoulli for each flag — all independent given the class:

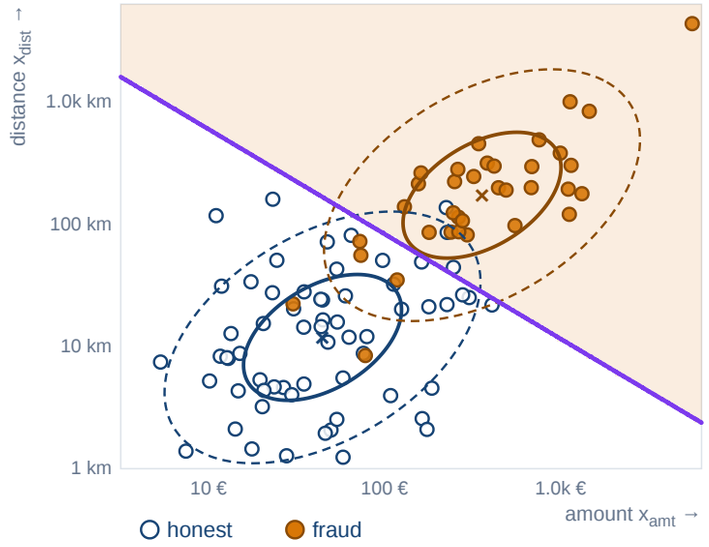
$$p(x \mid y;\theta) = \prod_{\text{numbers } j} \mathcal{N}(x_j;\ \mu_{jy}, \sigma^2_{jy}) \cdot \prod_{\text{flags } j} \theta_{jy}^{x_j}(1-\theta_{jy})^{1-x_j}$$

attribute of the transaction	fitted per class	likelihood ratio
amount 1,250 €	mean and spread of log amount	≈ 190
new device: yes	share of rows with the flag	6
both		$190 \cdot 6 = 1{,}140$

> Posterior odds $1{,}140 : 999$, so **53%** fraud. Fitting is still one pass over the table: a mean and a spread per number and class, a share per flag and class.

One Gaussian per Class

One shared covariance: the boundary becomes straight



One Σ shared by both classes

same spreads and correlation
for honest and fraud

per class: 2 means; Σ once: 3
boundary: a straight line

prior: the training share, 36% fraud

× mean — 1σ - - 2σ
violet: P(fraud | x) = 0.5

In fraud, amount and distance rise together. Which ellipses can follow the tilt — and what does one covariance for both classes do to the boundary?

Speaker notes

The 98 labelled transactions of Lecture 03, now with a normal fitted to each class's rows; the ellipses are the 1σ and 2σ contours, the violet line is where the fitted posterior is 0.5 (the prior is the training share of fraud), and the tinted region is decided as fraud.

Step 0, naive: amount and distance independent within each class, so each class gets a diagonal covariance and its ellipses stay upright; the model is the two-attribute version of the previous slide. Step 1, full: one covariance per class, which keeps the within-class correlation (0.78 in fraud in this sample); the fraud ellipse tilts with its cloud and the boundary bends with it. Step 2, shared: one covariance for both classes, estimated from all rows (the pooled within-class covariance); both ellipses have the same shape and the boundary is a straight line. The choice control returns to any of the three.

The Multivariate Normal

$$\mathcal{N}(x;\ \mu, \Sigma) = \frac{1}{(2\pi)^{p/2}\sqrt{\det \Sigma}} \exp\Big(-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big)$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\,\sigma_1\sigma_2 \\ \rho\,\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

The **covariance matrix** Σ: spreads on the diagonal, covariances off it.

model of $p(x \mid y)$ for p numbers	numbers per class	$p = 30$
counts, 10 bins per attribute	$10^p - 1$	10^{30}
Gaussian naive Bayes	$2p$	60
one Gaussian per class	$p + p(p+1)/2$	495

The family alone defeats the explosion: **quadratic** in p, not exponential — and the correlations stay in the model.

Speaker notes

In one dimension the formula is the normal density of Lecture 02. The exponent is minus one half of the squared Mahalanobis distance of x from μ; its level sets are the ellipses of the previous slide. The off-diagonal entries are the covariances of Lecture 02, $\mathrm{cov}(X_1, X_2) = \rho\sigma_1\sigma_2$. Σ is symmetric and must be positive definite; with fewer rows than attributes in a class its estimate is singular, which is one reason to share or to restrict it.

Counts: a mean vector has p entries, a symmetric $p \times p$ matrix $p(p+1)/2$ free ones (p variances and $p(p-1)/2$ covariances). For $p = 30$: $30 + 465 = 495$ per class. Binning instead needs a full joint over 10^{30} cells (Lecture 03). Gaussian naive Bayes is the special case of a diagonal Σ; a shared Σ needs p numbers per class plus $p(p+1)/2$ once.

The price of the family is its shape: one ellipse per class, symmetric, with a single peak. Where the data are shaped differently, the model is wrong in a way no amount of data repairs (the last slide of the module).

A Shared Covariance Gives a Linear Boundary

With the **same** Σ in both classes, the quadratic terms cancel in the log-odds:

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = \underbrace{(\mu_\text{f} - \mu_\text{h})^\top \Sigma^{-1}}_{w^\top} x + b \qquad \Longrightarrow \qquad P(\text{fraud} \mid x) = \frac{1}{1 + e^{-(w^\top x + b)}}$$

- **One Σ per class:** a quadratic boundary — *quadratic discriminant analysis*.
- **Shared Σ:** a straight boundary — *linear discriminant analysis*.
- **Diagonal Σ per class:** Gaussian naive Bayes, quadratic as well.

> A logistic function of a linear score: the form that logistic regression (Lecture 04) fits **directly**, without modelling x.

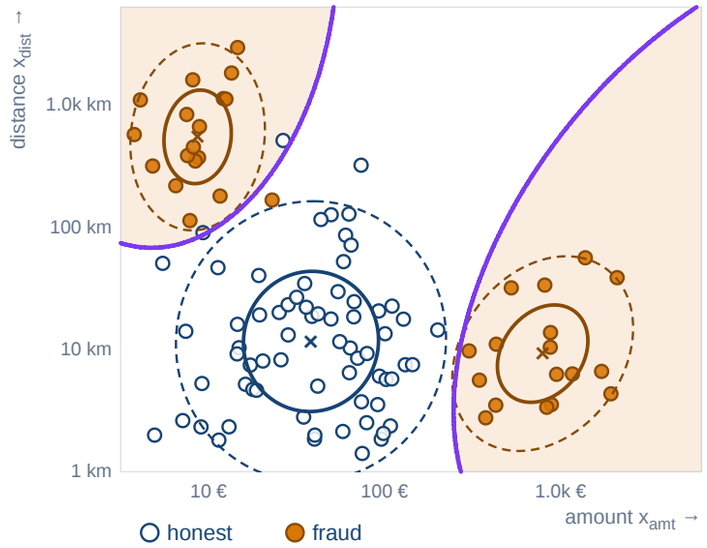
Speaker notes

Derivation. $\log \mathcal{N}(x;\mu,\Sigma) = -\tfrac{1}{2}x^\top\Sigma^{-1}x + \mu^\top\Sigma^{-1}x - \tfrac{1}{2}\mu^\top\Sigma^{-1}\mu + \text{const}$, where the constant depends on Σ only. In the difference of the two classes' logs the term $x^\top\Sigma^{-1}x$ and the constant cancel when Σ is shared, leaving $w = \Sigma^{-1}(\mu_\text{f} - \mu_\text{h})$ and $b = \tfrac{1}{2}\mu_\text{h}^\top\Sigma^{-1}\mu_\text{h} - \tfrac{1}{2}\mu_\text{f}^\top\Sigma^{-1}\mu_\text{f} + \log\frac{P(\text{fraud})}{P(\text{honest})}$. The prior enters only the offset b: changing the base rate shifts the line parallel to itself. With a separate Σ per class the quadratic term $-\tfrac{1}{2}x^\top(\Sigma_\text{f}^{-1} - \Sigma_\text{h}^{-1})x$ remains, and the boundary is a conic section (ellipse, parabola or hyperbola).

Posterior odds e^a correspond to the probability $1/(1+e^{-a})$, the logistic (sigmoid) function. This is Bishop, *Pattern Recognition and Machine Learning*, §4.2.1. Logistic regression fits w and b directly by maximising the likelihood of the labels; linear discriminant analysis obtains them from the class means, the shared covariance and the prior. The two estimates differ whenever the Gaussian assumption is wrong, the same contrast as naive Bayes against logistic regression for flags (Lecture 04).

When One Gaussian Is Not Enough

A mixture of two normals for fraud covers both patterns



Fraud: a mixture of two Gaussians

$p(x \mid \text{fraud}) = \pi_1 \, N(\mu_1, \Sigma_1) + \pi_2 \, N(\mu_2, \Sigma_2)$

$\pi = 0.51, 0.49$

fitted here with each pattern known;
in real data it is hidden

prior: the training share, 34% fraud

× mean — 1σ - - 2σ
violet: P(fraud | x) = 0.5

Fraud comes in two patterns. Where does a single normal put its mean — and what does that do to the boundary?

Speaker notes

Two kinds of fraud in one class: card testing, where stolen card data are tried with small amounts (around 9 €) far from the card holder's home, and large purchases (around 900 €) close to home with a stolen card. One normal for the fraud class has its mean between the two patterns, at an amount and distance where almost no fraud occurs, and a wide, tilted ellipse that spreads probability over the empty space between them. Its density is too low in both patterns and too high in the gap; the boundary follows the ellipse, not the data.

Step 1 fits a mixture of two normals for fraud, $p(x \mid \text{fraud}) = \pi_1 \mathcal{N}(x; \mu_1, \Sigma_1) + \pi_2 \mathcal{N}(x; \mu_2, \Sigma_2)$, with π_k the shares of the patterns. Here each row's pattern is known to the figure and the mixture is fitted pattern by pattern, by means and covariances. In real data the pattern is not recorded: it is a hidden variable, and fitting the mixture without it is the EM algorithm of Lecture 07. A model shape that is wrong does not improve with more rows; only a richer family does.

Mini-Exercise

Ten numeric attributes, two classes.

How many numbers does Gaussian naive Bayes need — and one full Gaussian per class? Count the prior too.

Speaker notes

Gaussian naive Bayes: per class 10 means and 10 variances, 20 numbers; two classes 40; the prior $P(\text{fraud})$ one more: 41. One Gaussian per class: 10 means and a symmetric 10×10 covariance matrix with $10 \cdot 11/2 = 55$ free entries, 65 per class, 130 for both, 131 with the prior. A shared covariance needs $2 \cdot 10 + 55 + 1 = 76$. Binned into 10 values each, the full joint would need $10^{10} - 1$ per class.

Mini-Exercise

Ten numeric attributes, two classes.

How many numbers does Gaussian naive Bayes need — and one full Gaussian per class? Count the prior too.

41 and 131. Naive: $10 + 10 = 20$ per class, $2 \cdot 20 + 1$. Full: $10 + 55 = 65$ per class, $2 \cdot 65 + 1$.

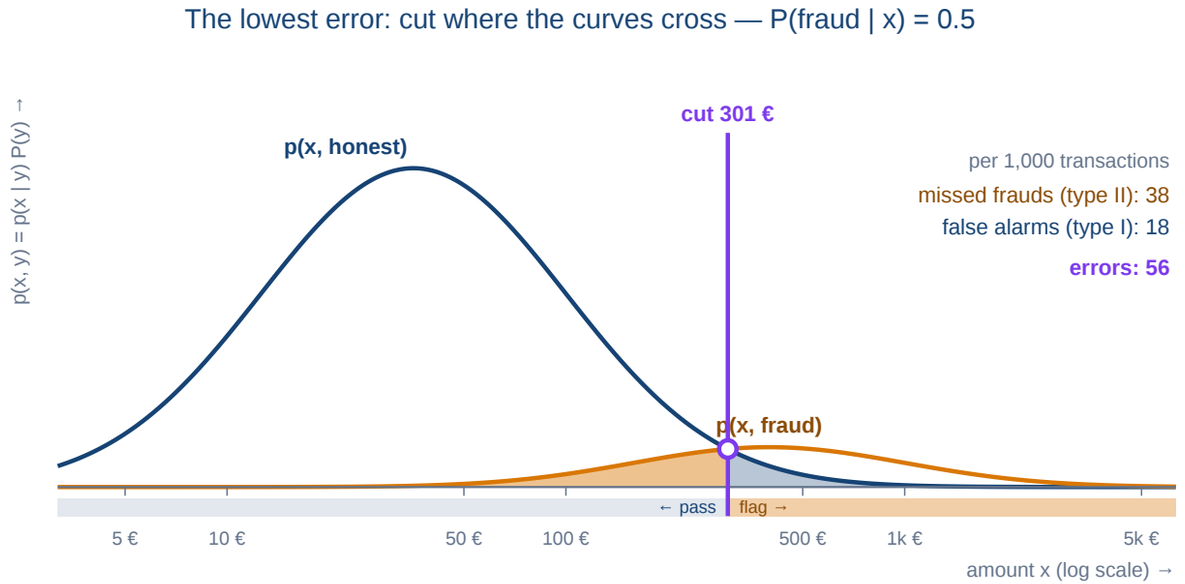
Takeaway

A number enters through a **density** per class; Bayes' theorem needs only the ratio of the two densities. **Gaussian naive Bayes** fits a mean and a spread per attribute and class; **one Gaussian per class** keeps the correlations at a quadratic, not exponential, cost; a **shared covariance** makes the boundary linear. One Gaussian cannot follow a class with two patterns — a **mixture** can.

Deciding on a Class

Guiding Question: The model says 17% fraud. Block, check or let it through — and who decides where the line is?

Decision Regions and Their Errors



Flag every transaction above the cut. Where should the cut go to make the fewest errors?

Speaker notes

The model of the previous module, one normal per class over $\log_{10}$ amount, with a fraud share of 10% so that the fraud curve is visible. Each curve is $p(x, y) = p(x \mid y)\, P(y)$: how much of all transactions belongs to that class at that amount; the area under each curve is its prior. A cut splits the axis into two **decision regions**, pass below and flag above. The frauds below the cut are passed (orange area), the honest transactions above it are flagged (navy area); the error rate is the sum of the two areas.

Step 1: a cut at 1,000 € passes 84 of 100 frauds per 1,000 transactions and flags 0.6 honest ones, 85 errors per 1,000. Step 2: between the cut and the point where the curves cross, the fraud curve lies above the honest one, so every transaction there is more often fraud than not and is passed anyway; that wedge (red) is avoidable. Step 3: the cut at the crossing, about 300 €, makes 38 + 18 = 56 errors per 1,000, the minimum. At the crossing $p(x \mid \text{fraud})P(\text{fraud}) = p(x \mid \text{honest})P(\text{honest})$, which is $P(\text{fraud} \mid x) = 0.5$. The slider and the button "best cut" repeat the experiment for other priors.

Two Kinds of Error

The best cut of the last slide, about 300 €, per 1,000 transactions (100 of them fraud):

	is fraud	is honest
flag (decide fraud)	62 — true positive	**18 — false positive**: type I error, a false alarm
pass (decide honest)	**38 — false negative**: type II error, a missed fraud	882 — true negative

Move the cut to 1,000 €. What happens to the two errors?

Speaker notes

"Positive" means flagged as fraud, the class searched for; "true" and "false" say whether the decision was right. The numbers follow from the two normals of the previous slide at the cut $\log_{10} 301$: $100 \cdot 0.381 = 38$ frauds lie below it, $900 \cdot 0.020 = 18$ honest transactions above it. As rates within each true class, the false positive rate is $18/900 = 2\%$ and the false negative rate $38/100 = 38\%$; one minus the latter, 62%, is the share of frauds caught (recall, in Lecture 06).

The names type I and type II come from statistical testing. Read the decision as a test of the null hypothesis H_0: "the transaction is honest". A type I error rejects H_0 although it is true (an honest transaction flagged); its probability is the significance level α of a test. A type II error keeps H_0 although it is false (a fraud let through); its probability is β, and $1 - \beta$ is the power. For a given model, α and β move in opposite directions with the cut: at 100 € the false alarms rise to 143 and the missed frauds fall to 7 (150 errors); at 1,000 € they are 0.6 and 84 (85 errors). Only better attributes, which separate the two curves further, lower both at once.

Two Kinds of Error

The best cut of the last slide, about 300 €, per 1,000 transactions (100 of them fraud):

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Move the cut to 1,000 €. What happens to the two errors?

> False alarms fall to 0.6, missed frauds rise to 84: moving the cut **trades one error for the other**. The Bayes rule minimises their **sum**; costs, next, weigh them.

The Bayes Decision Rule

The **Bayes decision rule** decides the most probable class:

$$\hat{y}(x) = \arg\max_y P(y \mid x) = \arg\max_y p(x \mid y)\, P(y)$$

No other rule on the same attributes makes fewer errors.

Why it errs least

At each x, deciding $\hat{y}$ is wrong with probability $1 - P(\hat{y} \mid x)$ — smallest for the most probable class. Best at every x is best on average.

Two classes, as odds

Decide fraud iff $P(\text{fraud} \mid x) > 0.5$, i.e.

$$\text{LR}(x) > \frac{P(\text{honest})}{P(\text{fraud})} = 999$$

Speaker notes

The error rate of a rule $\hat{y}$ is $P(\text{error}) = \int \big(1 - P(\hat{y}(x) \mid x)\big)\, p(x)\, dx$. The integrand is non-negative and depends on the rule only through the class it picks at x, so the integral is smallest when every integrand is smallest, which is the class with the largest posterior. Since $p(x)$ does not depend on y, the largest posterior is the largest $p(x \mid y)P(y)$: the higher curve of the previous slide. The regions where each class is decided are the decision regions; the boundary between them is where the posteriors are equal. The same rule holds for any number of classes.

What remains is the area under the lower curve everywhere, the overlap of the classes. It is the Bayes error of Lecture 06: the lowest error any classifier can reach with these attributes, a property of the data, not of the model. The rule is also called the MAP decision (maximum a posteriori class). It should not be confused with MAP *estimation* (Lecture 04), which picks the most probable parameter value given the training data; both maximise a posterior, over different unknowns.

Odds form, from Lecture 03: the posterior odds are the prior odds times the likelihood ratio, and they exceed 1 iff $\text{LR}(x) > P(\text{honest})/P(\text{fraud})$. At the base rate of 1 in 1,000 the three red flags (ratio 210) and the amount of 1,250 € (190) each fall short; amount and new device together (1,140) pass it.

Unequal Costs

	is fraud	is honest
check it (a text to the customer)	0	€5 (c_{FP}, false positive)
let it through	€500 (c_{FN}, false negative)	0

With $p = P(\text{fraud} \mid x)$, at which p is checking the cheaper action?

Speaker notes

The table is a cost matrix: the cost of each action in each true state. The expected cost of an action is its cost in each state weighted by the posterior: checking costs c_{FP} when the transaction is honest, letting it through costs c_{FN} when it is fraud. The cheaper action is chosen at every x; with zero cost for correct actions the rule is $p > c_{FP}/(c_{FP} + c_{FN})$. Only the ratio of the costs matters. Equal costs give $p^* = 0.5$, the Bayes decision rule of the previous slide, which therefore minimises the expected cost exactly when both errors cost the same.

In general, with actions a and costs $c(a, y)$, the rule chooses $\arg\min_a \sum_y c(a, y)\, P(y \mid x)$, the action with the smallest expected cost (the Bayes risk at x). A third action, deferring to a human, adds a band of posteriors around the threshold in which neither automatic action is cheap enough (Lecture 01).

The three-flag transaction of Lecture 03 has posterior 17%: checking costs 0.83 × €5 = €4.15 in expectation, letting it through 0.17 × €500 = €85. The threshold applies to a calibrated probability; the naive Bayes posterior is over-confident where flags are correlated, so the threshold is applied to a number that may be too high (Lecture 04).

Unequal Costs

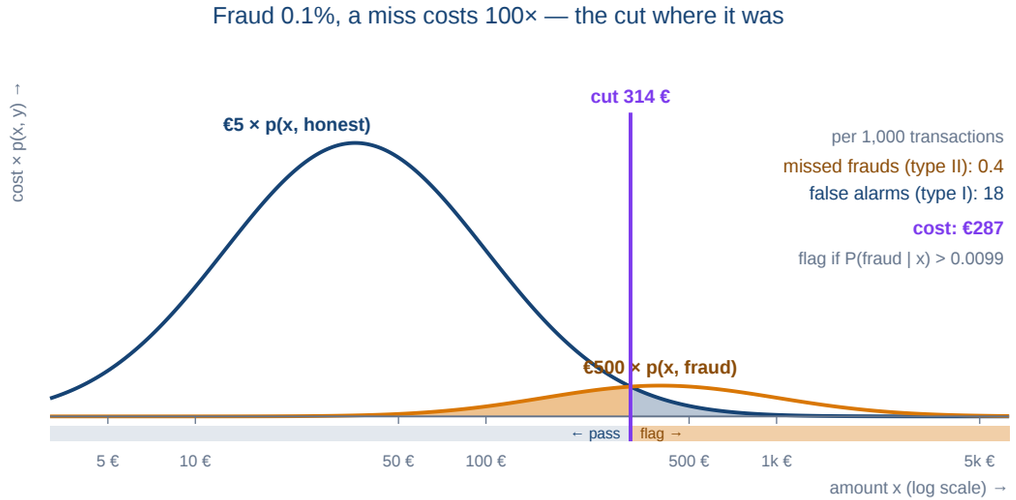
	is fraud	is honest
check it (a text to the customer)	0	€5 (c_{FP}, false positive)
let it through	€500 (c_{FN}, false negative)	0

With $p = P(\text{fraud} \mid x)$, at which p is checking the cheaper action?

$$\underbrace{(1-p)\,c_{FP}}_{\text{check}} < \underbrace{p\,c_{FN}}_{\text{let through}} \quad \Longleftrightarrow \quad p > p^* = \frac{c_{FP}}{c_{FP}+c_{FN}} = \frac{5}{505} \approx 1\%$$

> Lecture 01's loan threshold, for fraud. The three-flag transaction (17%) is not more likely fraud than not — and well worth a check.

Costs and Prior in One Threshold



$$\text{check iff} \quad \text{LR}(x) > \frac{c_{FP}}{c_{FN}} \cdot \frac{P(\text{honest})}{P(\text{fraud})}$$

Predict before the last click: fraud becomes 100 times rarer. Where does the cut go?

Speaker notes

In odds form the rule $p > c_{FP}/(c_{FP} + c_{FN})$ reads: posterior odds $> c_{FP}/c_{FN}$, and with posterior odds = prior odds × likelihood ratio, $\mathrm{LR}(x) > (c_{FP}/c_{FN}) \cdot P(\mathrm{honest})/P(\mathrm{fraud})$. The costs and the prior enter only as this product; the model's part, the likelihood ratio, is untouched. In the figure each curve is weighted by the cost of misclassifying its class, $c_{FN}\,p(x, \mathrm{fraud})$ and $c_{FP}\,p(x, \mathrm{honest})$, and the best cut is where the weighted curves cross.

Step 0: fraud share 10%, equal costs: threshold on the ratio 9, cut at about 300 €. Step 1: a missed fraud costs €500, a check €5: the fraud curve grows a hundredfold, the threshold on the ratio falls to 0.09 and the cut moves to about 49 €, flagging more than a third of the honest transactions — at a fraud share of 10% that is the cheaper policy. Step 2: fraud share 0.1%: the threshold is $0.01 \cdot 999 \approx 10$ again and the cut returns to about 314 €. A class that is 100 times rarer and 100 times costlier to miss gets the same boundary. At the real base rate with equal costs the cut would lie at about 3,000 €.

A generative model changes both factors without refitting: a new base rate replaces $P(y)$, new costs replace c_{FP}/c_{FN}. A discriminative model can take new costs the same way, and a new base rate by correcting its odds (Lecture 04); a rule has to be learned again. How the threshold trades recall against precision, and how its cost curve looks, is Lecture 06.

Mini-Exercise

A transaction has likelihood ratio **20**; prior odds $1 : 999$.

Is it fraud under the Bayes decision rule? Is it worth a €5 check against a €500 loss?

Speaker notes

Posterior odds $20 : 999$, probability $20/1{,}019 = 0.0196$. Under the Bayes decision rule (equal costs) fraud would need a ratio above 999. With c_{FP} = €5 and c_{FN} = €500 the threshold on the posterior is $5/505 = 0.0099$, on the likelihood ratio $(5/500) \cdot 999 = 9.99$; the ratio 20 exceeds it. Expected costs: checking 0.98 × €5 = €4.90, letting it through 0.0196 × €500 = €9.80.

Mini-Exercise

A transaction has likelihood ratio **20**; prior odds $1 : 999$.

Is it fraud under the Bayes decision rule? Is it worth a €5 check against a €500 loss?

No, and yes. $20 < 999$: $P(\text{fraud} \mid x) = 20/1{,}019 \approx 2\%$. The check pays above 1%: on the ratio, above $999 \cdot 5/500 \approx 10$.

Takeaway

Deciding the **most probable class** gives the fewest errors — the **Bayes decision rule**; what remains is the overlap of the classes. **Unequal costs** move the threshold on the posterior to $c_{FP}/(c_{FP} + c_{FN})$. For a generative model costs and **prior** meet in one threshold on the likelihood ratio: both change without refitting $p(x \mid y)$.